

UNOFFICIAL ENGLISH TRANSLATION

Trophodynamics of marine organisms in the epipelagic layer of the Okhotsk Sea in the 2000s

K. M. Gorbatenko and I. V. Melnikov

Izvestiya TINRO (2019), volume 198, pages 143-163. DOI: 10.26428/1606-9919-2019-198-143-163.

Prepared for research reading on 3 October 2026. This is an unofficial AI-assisted translation from Russian, not an author-approved or publisher-issued English edition. The original PDF is authoritative. All main text, three tables, ten figures and their captions, author statements, both bibliographies, and publication dates are included. The existing English abstract and English reference list are retained, with typographic normalization.

Each section and footer identifies the original printed page and PDF page. The English translation uses a new page layout and is longer than the 21-page source. Sentence continuations across original page boundaries are marked by ellipses. Original figure graphics retain their numerical values and Russian labels; accompanying English keys translate all labels and legends. Figure 9 is reproduced from its original embedded image and has a complete English node key.

Printed numerical and scientific inconsistencies are preserved rather than corrected. A small number of clearly marked translation notes explain apparent source inconsistencies or wording differences. In particular, the published English abstract gives annual nekton consumption in tC, while the Russian abstract gives tonnes of food. The two abstracts are both retained.

Source PDF: <https://izvestiya.tinro-center.ru/jour/article/download/497/467>

Source SHA-256: 70b73d4341018e3fadb9955e9d19735188f12c8798eaed8921d1812bb6c2b860

Citation: Gorbatenko, K. M., and Melnikov, I. V. (2019). Trophodynamics of marine organisms in the epipelagic layer of the Okhotsk Sea in 2000s. *Izvestiya TINRO*, 198, 143-163. <https://doi.org/10.26428/1606-9919-2019-198-143-163>

Original page 143 (source PDF page 1)

Trophodynamics of marine organisms in the epipelagic layer of the Okhotsk Sea in the 2000s

K. M. Gorbatenko and I. V. Melnikov

Pacific Branch of VNIRO (TINRO), 4 Shevchenko Alley, Vladivostok 690091, Russia.

Izvestiya TINRO, 2019, volume 198, pp. 143-163. UDC 574.523(265.53).

DOI: 10.26428/1606-9919-2019-198-143-163.

Abstract translated from Russian

Trophodynamic modeling provided new data on the transfer of matter and energy among the principal biotic components of the Okhotsk Sea ecosystem, taking into account their rates of production and food consumption. Research data on zooplankton and nekton, the feeding of aquatic organisms, isotope composition, and organic carbon made it possible to substantiate the principal trophodynamic relationships in pelagic and bottom communities of the Okhotsk Sea for 2000-2014. Total zooplankton production for the entire sea was 2616 million tonnes wet weight (non-predatory plankton, 2275 million tonnes; predatory plankton, 341 million tonnes). In addition to favorable climatic and oceanographic conditions, the high total zooplankton production of the Okhotsk Sea is associated with the dominance of highly productive species. Calculations of zooplankton consumption by predators showed that only 22.4% of total annual zooplankton production is consumed during the year (predatory plankton, 16.2%; nekton, 6.2%). Using organic carbon as a measure of energy made it possible to establish that annual organic matter production in the Okhotsk Sea pelagic zone is 831.0 million tC at trophic level I, 177.400 at II, 18.100 at III, 0.740 at IV, and 0.016 million tC/year at V. Comparing production and consumption levels indicates that pelagic nekton consumed an average of 159 million tonnes of food per year in the 2000s (2000-2014). In the annual nekton diet, zooplankton accounted for 85.5%, nekton for 12.8%, and zoobenthos for 1.7% by biomass. Pollock accounted for 50.7% of total nekton food consumption in the Okhotsk Sea epipelagic layer; herring, 18.9%; squid, 16.6%; capelin, 7.6%; deep-sea smelt, 5.3%; and salmon, 0.9%. In total, nekton consumes only 6.2% of annual zooplankton production. Total annual production of aquatic organisms in the epipelagic layer exceeds 1 billion tC (1027.4 million tC/year, or 17.85 billion tonnes wet weight). Excluding primary production, which constitutes 67.60% of gross production in carbon equivalents, the largest contributions to production are made by microheterotrophs, 13.30%, and the dominant zooplankton groups (copepods, 11.40%; euphausiids, 5.50%; sagittas, 1.20%; hyperiids, 0.50%); nekton accounts for only 0.13%.

Keywords: trophodynamic relationships; epipelagic layer; zooplankton; nekton; Okhotsk Sea.

* Konstantin Mikhailovich Gorbatenko, Doctor of Biological Sciences, head of section, email: konstantin.gorbatenko@tinro-center.ru; Igor Vladimirovich Melnikov, Candidate of Biological Sciences, deputy director, email: igor.melnikov@tinro-center.ru.

The source also supplies these author details in English: Gorbatenko Konstantin M., D.Biol., head of section, Pacific Branch of VNIRO (TINRO), Shevchenko Alley, 4, Vladivostok, 690091, Russia; Melnikov Igor V., Ph.D., deputy director, at the same address. Email addresses are as above.

Original page 144 (source PDF page 2)

English abstract as published in the source

Gorbatenko K.M., Melnikov I.V. Trophodynamics of marine organisms in the epipelagic layer of the Okhotsk Sea in 2000s // Izv. TINRO. - 2019. - Vol. 198. - P. 143-163.

New data on matter and energy transfer between major components of the Okhotsk Sea ecosystem are obtained on the base of trophodynamic modeling, taking into consideration their production and food consumption rates. The main trophodynamic relationships in the pelagic and bottom communities are determined from observations on zooplankton and nekton abundance, organic carbon content, food habits of marine organisms, and their isotope composition in 2000-2014. The total zooplankton production in the entire Okhotsk Sea in these years is assessed as 2616×10^6 t in raw weight, including 2275×10^6 t for non-predatory plankton, and 341×10^6 t for predatory plankton. So high total production of zooplankton is conditioned by favorable environmental conditions and dominance of high-productive species. Taking into account the rate of zooplankton consumption by predators, only 22.4% of the total annual zooplankton production was consumed annually, with 16.2% grazed by predatory plankton and 6.2% by nekton. In carbon units, 831.0×10^6 tC was produced annually in the Okhotsk Sea at the first trophic level, 177.4×10^6 tC at the second trophic level, 18.1×10^6 tC at the third trophic level, 0.74×10^6 tC at the fourth trophic level, and 0.016×10^6 tC at the fifth trophic level. Pelagic nekton consumed 159×10^6 tC annually. The nekton prey included 85.5% of zooplankton, 12.8% of nekton, and 1.7% of zoobenthos, by biomass. The main part of zooplankton consumed by nekton (50.7%) was grazed by walleye pollock, 18.9% by herring, 16.6% by squids, 7.6% by capelin, 5.3% by deep-sea smelt, and 0.9% by salmons. The total annual production of organisms in the epipelagic layer of the Okhotsk Sea exceeded 10^9 tons of C (1027.4×10^6 tC/year equal to the biomass of 17.85×10^9 t in wet weight). Primary production is estimated as 67.60% of gross production in carbon units, microheterotrophic organisms produce 13.30%, dominant zooplankton groups - 18.60% (copepods 11.40%, euphausiids 5.50%, sagittas 1.20%, and hyperiids 0.50%), the portion of nekton production is estimated as 0.13% of gross production.

Kew words: trophodynamic link, epipelagic layer, zooplankton, nekton, Okhotsk Sea.

Introduction

The Okhotsk Sea is Russia's principal fishing region, and the condition of its biological resources has been monitored regularly for many decades [Shuntov, 2016]. The study of the food supply and trophic relationships of aquatic organisms has almost always formed part of this research. With the practical introduction of ecological monitoring and ecosystem approaches, trophological research has needed to expand and deepen, including clarification of the trophic status and energetic value of aquatic organisms.

One of the principal tasks in research on the trophodynamics of aquatic organisms is to establish the patterns of biological transformation of matter and energy in water bodies. In aquatic ecosystems, identifying discrete trophic levels between which matter and energy are transferred is difficult because a substantial part of plant and animal production accumulates as detritus. This is consumed by many aquatic species, particularly invertebrates, a large proportion of which are omnivorous. Identifying and assessing energy flows in water bodies therefore requires determination of the actual trophic status and energy potential of all abundant species constituting the food web.

Until now, the study of aquatic ecosystems through analysis of natural stable-isotope ratios and determination of the quantity of organic carbon in aquatic organisms has not been widely used in Russian hydrobiological research. Given the importance of this problem, such studies were initiated at TINRO-Center in 2003 [Gorbatenko et al., 2004a].

One direction of trophological research is ecosystem modeling, which is becoming increasingly developed in studies of many regions of the World Ocean, including the Far Eastern seas. Trophodynamic models previously constructed for these seas [Dulepova, 2002; Radchenko, 2011, 2015; Zavolokin, 2014] made it possible, at a certain stage, to investigate ecosystem process dynamics and their balance. The models presented differed substantially...

Original page 145 (source PDF page 3)

...because complete information on the abundance of all ecosystem components, particularly those at the lower trophic levels, and their roles in trophic flows was lacking [Zavolokin, 2014], as were reliable energy indices for aquatic organisms expressed in calories or organic carbon. Earlier, in constructing trophodynamic models, V. I. Radchenko [2011] noted that at the present stage of research on trophic relationships of Okhotsk Sea plankton, we cannot provide reliable information on diets or determine what proportions of the diets of individual species, or even groups of plankton, consist of autotrophs and heterotrophs.

There are several approaches to determining primary production and energy transformation in food webs, but modern studies most often express quantitative indices in units of carbon mass. Carbon is the principal element of the organic matter transformed in ecosystems and occurs in every component of organisms: fats, proteins, and carbohydrates. Studies of the formation of organic carbon and its transformation pathways require its quantity in aquatic organisms to be determined so that subsequent matter or energy flows in the food web can be expressed in units of carbon mass. At present, the chemical composition of organisms is most easily and accurately determined by combustion in an elemental analyzer (a CN analyzer).

This study uses information obtained during TINRO-Center expeditions in the 2000s that investigated the Okhotsk Sea epipelagic layer: the composition, structure, and interannual dynamics of plankton and nekton communities; the feeding of aquatic organisms; the isotope composition of soft tissues; and the quantity of organic carbon in dominant populations.

The principal objective is a quantitative assessment of the trophodynamics of pelagic aquatic organisms in the epipelagic layer of the Okhotsk Sea.

Materials and methods

Plankton was sampled in the 0-200 m layer (surface to bottom at depths below 200 m) with a BSD net (mouth area 0.1 m², mesh size 0.168 mm), and processed using the standard methodology adopted at TINRO-Center, incorporating corrections for under-sampling of individual plankton groups [Volkov, 2008]. More than 10,000 plankton samples were collected and processed.

Daily, seasonal, and annual food consumption had previously been determined for various planktonic and nektonic organisms forming the basis of pelagic biocoenoses; the materials and methods have already been described in detail in the literature [Gorbatenko et al., 2004a, b; Gorbatenko, 2016a, b; Gorbatenko and Levitskaya, 2016; Gorbatenko and Melnikov, 2016].

Zooplankton and nekton were collected for isotope analysis and organic carbon determination in 2003-2014. Net plankton was collected with Bongo, BSD, and IKS plankton nets. Plankton stations and trawl stations coincided (Fig. 1). In most investigations of the pelagic zone, an 80/376 midwater trawl with a small-mesh liner was used.

The ratios of stable carbon isotopes (¹³C and ¹²C) and nitrogen isotopes (¹⁵N and ¹⁴N) in the samples were determined in the conventional form as the relative quantity δ , in per mille, measured as the difference between the isotope ratio in the sample and in a standard:

For carbon: $\delta^{13}\text{C} (\text{‰}) = [({}^{13}\text{C}/{}^{12}\text{C}_{\text{sample}} - {}^{13}\text{C}/{}^{12}\text{C}_{\text{standard}})/({}^{13}\text{C}/{}^{12}\text{C}_{\text{standard}})] \times 1000$.

For nitrogen: $\delta^{15}\text{N} (\text{‰}) = [({}^{15}\text{N}/{}^{14}\text{N}_{\text{sample}} - {}^{15}\text{N}/{}^{14}\text{N}_{\text{standard}})/({}^{15}\text{N}/{}^{14}\text{N}_{\text{standard}})] \times 1000$.

To standardize the data, stable carbon isotope values are expressed relative to Vienna Pee Dee Belemnite carbonate (V-PDB), and nitrogen values relative to atmospheric nitrogen. Calibration used IAEA CH-6, NBS-22, IAEA N-1, and IAEA N-2 standards (International Atomic Energy Agency, Vienna). The precision of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ determinations was ± 0.06 and $\pm 0.10\text{‰}$, respectively.

Original page 146 (source PDF page 4)

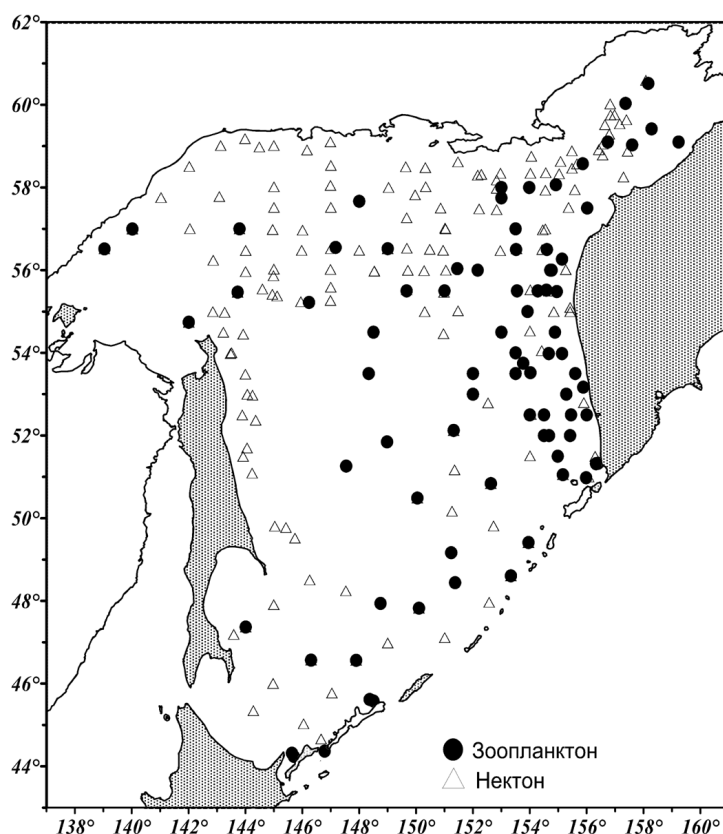


Figure 1. Stations at which aquatic organisms were sampled in the Okhotsk Sea in 2003-2014 to determine stable-isotope ratios and the quantity of organic carbon in their tissues.

English figure key: The map is reproduced unchanged. Solid black circles: zooplankton. Open triangles: nekton. Sampling coordinates and their distribution are retained as in the original.

The actual trophic level occupied by a species in aquatic food webs was determined from nitrogen isotope composition [Post, 2002]:

$$TL_c = \lambda + (\delta^{15}N_c - \delta^{15}N_b) / \Delta.$$

Here TL_c is the trophic level of the consumer; $\delta^{15}N_c$ is the nitrogen isotope composition of the consumer; $\delta^{15}N_b$ is the nitrogen isotope composition of organisms used as the base of the food chain (primary producers or primary consumers); Δ is the change in isotope composition over one trophic level, taken as 3.4‰ [Minagawa and Wada, 1984]; and λ is the trophic level of the organisms taken as the base of the food chain (I for primary producers or II for primary consumers).

Because direct measurements of the isotope composition of phytoplankton were unavailable, the copepod *Eucalanus bungii*, a first-order consumer, was used as the base of the pelagic food chain. This copepod species, a fine-particle filter feeder, showed the lowest $\delta^{15}N$ values among Okhotsk Sea zooplankton in every season in our data. We therefore consider it to occupy trophic level II [Gorbatenko et al., 2014]. We also adopted $\delta^{15}N_b$ values of 7.8 in spring, 6.9 in summer, 7.8 in autumn, and an overall seasonal mean of 7.4.

To determine organic carbon and nitrogen, samples of individual aquatic species consisted of 1-100 individuals, depending on organism size. For nekton, 3-10 g of muscle tissue was taken from the dorsal side, rinsed with distilled water to remove salts, and dried in an oven for 12-24 hours at 60°C until fully dehydrated. The dried samples were ground with a corundum mortar and pestle, redried to constant mass, and used to determine organic carbon and nitrogen by combustion in a EURO-3000 elemental CNH analyzer.

A review of production calculation methods showed that all are based on data on somatic and reproductive growth of the individuals constituting a population [Zaika, 1972; Sushchenya, 1975; Vinberg, 1979; Ivanova, 1985; and others]. Determining population production requires information on the rate of production by individuals, that is, growth rate,...

Original page 147 (source PDF page 5)

...the formation of reproductive products and exuviae, and the excretion of liquid metabolites [Ivanova, 1985]. Data on age composition, biomass, and density were used to calculate the specific daily production of aquatic organisms. Copepod production was calculated conventionally using the Boysen-Jensen equation [Boysen-Jensen, 1919]:

$$P_t = B_2 - B_1 + B_e.$$

Here B_1 and B_2 are species biomass at the beginning and end of the observation period $t = t_2 - t_1$; B_e is the loss due to grazing, natural mortality, and losses of matter during life. B_e was calculated from:

$$B_e = N_1 - N_2 (1/2)(B_1/N_1 + B_2/N_2).$$

Here N_1 and N_2 are the initial and final abundance, respectively, over a specified period.

Translation note: The grouping of N_1 and N_2 in this displayed equation follows the printed source. No missing parentheses have been supplied.

Production of systems above the population level, comprising several trophic levels, was determined using conventional methods [Zaika, 1983; Ivanova, 1985; Vinogradov and Shushkina, 1987]. Production at a trophic level equals the sum of production by its constituent populations. Production of a system consisting of n trophic levels is:

$$P = P_1 + P_2 + \dots + P_n - A_2 - \dots - A_n.$$

Here P_n and A_n are the production and assimilated energy, respectively, at trophic level n .

Total production of a zooplankton community comprises the summed production of non-predatory zooplankton (phytophagous and euryphagous forms) and predatory zooplankton, minus the food assimilated by predators [Zaika, 1983; Ivanova, 1985]. Total zooplankton community production is determined by:

$$P(\text{zooplankton}) = P(\text{non-predatory zooplankton}) + P(\text{predatory zooplankton}) - P(\text{predatory zooplankton})/K.$$

K is the second-order efficiency coefficient for predatory plankton. According to Yu. I. Sorokin [1982], K for predatory plankton in marine ecosystems is 0.3-0.4. Total production indicates the overall volume of organic matter generated by zooplankton. Hydrobiological studies assessing food resources also use the actual production of zooplankton, calculated as the difference between the summed production of all community components and the part consumed within the community by planktonic predators. Actual production is thus the portion of total production available to fish as food.

Production characteristics of Okhotsk Sea zooplankton had previously been studied in dominant species [Shebanova, 2007; Shebanova et al., 2010, 2011, 2012; Chuchukalo et al., 2013]. The data in these studies made it possible to calculate the production of abundant species forming the biomass basis of the four dominant taxonomic groups of Okhotsk Sea zooplankton.

Results and discussion

Plankton

On the basis of published data [Shushkina and Vinogradov, 1988; Sorokin et al., 1995, 1997; Naletova et al., 1997; Matveev, 2006; Matveev and Zhigalov, 2008], obtained principally during academic research, we presently adopt Okhotsk Sea primary production, including recycling, as $450 \text{ gC/m}^2/\text{year}$, bacterial production as $67.5 \text{ gC/m}^2/\text{year}$, and protozoan production as $27.0 \text{ gC/m}^2/\text{year}$. Together with phytoplankton, the microheterotrophs comprising suspended organic matter form the marine foundation on which all trophic pyramids are based [Sorokin, 1977].

Zooplankton studies at TINRO-Center over a 30-year period showed that five species dominated throughout the Okhotsk Sea all year. These are the interzonal...

Original page 148 (source PDF page 6)

...copepods *Metridia okhotensis* and *Neocalanus plumchrus*, the euphausiids *Thysanoessa raschii* and *Thysanoessa longipes*, and the sagitta *Sagitta elegans* [Gorbatenko, 1997]. Calculations of mean zooplankton biomass in the 2000s produced values close to the long-term means for the entire study period (Fig. 2). Long-term mean zooplankton biomass in the epipelagic layer during the 2000s is estimated at 214.6 million tonnes wet weight, or 140.9 g/m². Copepods account for 48.3% of biomass; euphausiids, 26.9%; sagittas, 19.9%; and amphipods, 2.7%.

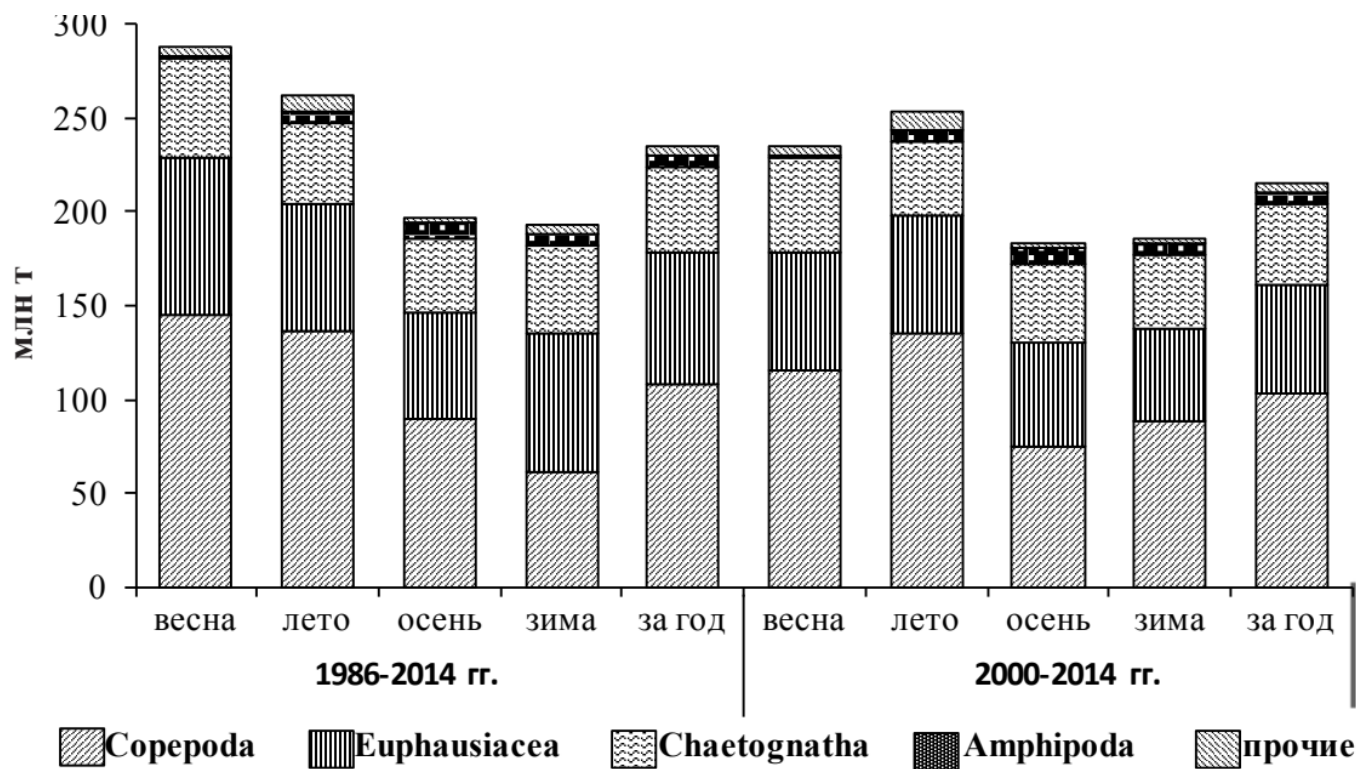


Figure 2. Long-term mean zooplankton biomass in the Okhotsk Sea in different study periods.

English figure key: Vertical axis: million tonnes. Horizontal axis, in both periods: spring; summer; autumn; winter; annual mean. Left period: 1986-2014; right period: 2000-2014. Legend, left to right: Copepoda; Euphausiacea; Chaetognatha; Amphipoda; other.

Table 1 presents the mean biomass and production of the principal dominant zooplankton groups in the Okhotsk Sea by season.

Table 1. Long-term mean (2000-2014) biomass (B) and production (P) of dominant plankton groups in the epipelagic layer of the Okhotsk Sea, million tonnes.

Group	Winter			Spring			Summer			Autumn		
	B	P	P/B	B	P	P/B	B	P	P/B	B	P	P/B
Euphausiids	49.4	98.8	2.0	61.4	141.2	2.3	63.8	210.5	3.3	56.0	142.2	2.5
Copepods	88.2	245.2	2.8	116.5	435.7	3.7	135.4	705.4	5.2	74.1	259.4	3.5
Hyperiid	6.4	15.3	2.4	1.6	5.3	3.3	6.9	19.3	2.8	8.7	28.3	3.3
Sagittas	39.3	27.5	0.7	51.2	71.7	1.4	37.8	94.5	2.5	42.5	59.5	1.4
Other	3.1	6.1	2.0	4.8	12.9	2.7	9.0	31.2	3.5	2.2	5.8	2.7
Total	186.4	392.9	2.1	235.5	666.8	2.8	252.9	1060.9	4.2	183.5	495.2	2.7

Note: $\pm$ SE did not exceed 10% of the biomasses presented.

Calculations were made for different biotopes (coastal zone, shelf waters, and open waters) by season; weighted means then yielded average production for individual seasons throughout the sea. Copepods have the highest production: total winter production was estimated at 245.2 million tonnes (P/B = 2.8), spring at 435.7 (P/B = 3.7), summer at 705.4 (P/B = 5.2), and autumn at 259.4 million tonnes (P/B = 3.5) (Table 1). Annual copepod production was estimated at 1645.7 million tonnes wet weight, and the mean seasonal P/B ratio is 3.8. The high production...

Original page 149 (source PDF page 7)

...of copepods in the Okhotsk Sea (annual P/B = 15.2) is caused by the dominance of two species, *N. plumchrus* and *M. okhotensis* [Gorbatenko, 1997], which have maximal production and annual P/B ratios above 15 [Shebanova, 2007; Shebanova et al., 2010, 2011, 2012; Chuchukalo et al., 2013]. In contrast, the dominant copepods of the Bering Sea and Pacific subarctic waters (the northwestern Pacific, NWP) include less productive species: *Neocalanus cristatus*, *Metridia pacifica*, and *E. bungii*, whose annual P/B ratios are below 10. The two most productive euphausiid species, *T. longipes* and *T. raschii*, also predominate absolutely in the Okhotsk Sea [Gorbatenko, 2018], with annual P/B ratios above 10 [Chuchukalo et al., 2013]. Compared with the NWP, where *Euphausia pacifica* dominates with an annual P/B ratio of 6, euphausiid production in the Okhotsk Sea is more than 1.5 times that in adjacent Pacific waters. Among hyperiids, the most productive species, *Themisto pacifica*, also dominates in the Okhotsk Sea [Shebanova et al., 2014]. Thus the Okhotsk Sea can be considered the most productive water body because it is dominated by zooplankton species with maximal production.

Total epipelagic zooplankton production in the 2000s was estimated at 392.9 million tonnes in winter (P/B = 2.1), 666.8 in spring (P/B = 2.8), 1060.9 in summer (P/B = 4.2), and 495.2 million tonnes in autumn (P/B = 2.7) (Table 1).

Within zooplankton communities, low-trophic-status plankters, phytophages and euryphages, play the principal role in generating organic matter. The role of planktonic predators is also very substantial: on the one hand they generate organic matter, and on the other they consume filter feeders and reduce their production. Mean total annual zooplankton production for the entire sea in the 2000s was estimated at 2616 million tonnes wet weight, including 2275 million tonnes of non-predatory plankton, mainly copepods and euphausiids, and 341 million tonnes of predatory plankton, mainly sagittas and hyperiids.

Feeding

Food composition and daily ration data for non-predatory and predatory plankton [Gorbatenko, 2016a, b] and nekton [Gorbatenko and Levitskaya, 2016; Gorbatenko and Melnikov, 2016] in the Okhotsk Sea during the 2000s were used to calculate prey consumption in different seasons and throughout the year.

Feeding data for predatory zooplankton, principally sagittas [Gorbatenko, 2016b], showed that predatory plankton consumes a total of 424 million tonnes annually. This is about 50.5% of the gross stock, or 16.2% of zooplankton production. The principal pressure falls on copepod populations: 19.7% of that group's total annual production (Table 2).

Table 2. Shares of annual zooplankton production and that of its dominant groups consumed by predatory plankton and nekton in the epipelagic layer of the Okhotsk Sea, %.

Food component	Predatory plankton	Nekton	Total
All zooplankton	16.2	6.2	22.4
Euphausiids	1.3	15.7	17.0
Copepods	19.7	2.7	22.4
Hyperiids	0.2	26.0	26.2
Sagittas	1.8	1.1	2.9

Nekton prey consumption estimates established that its annual ration of 195 million tonnes [Gorbatenko and Levitskaya,...

Original page 150 (source PDF page 8)

...2016] requires 15.7% of annual euphausiid production, 2.7% of copepod production, 26.0% of hyperiid production, and 1.1% of sagitta production (Table 2). Pollock is the principal consumer (Fig. 3).

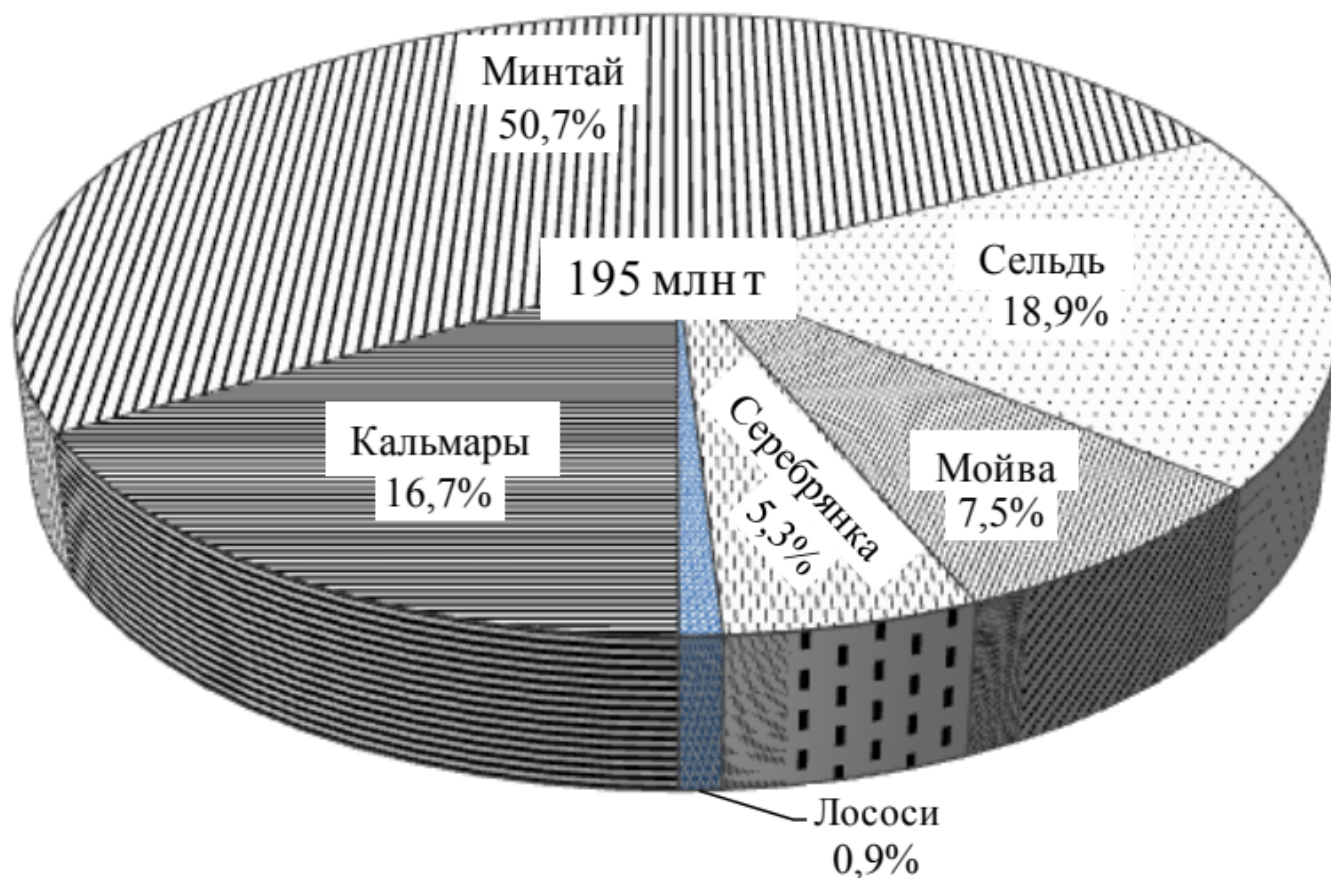


Figure 3. Long-term mean annual food consumption by dominant nekton species, 2000-2014.

English figure key: Pollock 50.7%; herring 18.9%; capelin 7.5%; squid 16.7%; deep-sea smelt 5.3%; salmon 0.9%. Total: 195 million tonnes. Values retain the rounding printed in the figure.

Thus the greatest pressure from nekton was on hyperiids and euphausiids, the plankton groups least consumed by predatory plankton (Table 2). Throughout the year, predatory zooplankton and nekton together consumed 22.4% of total annual zooplankton production. Pressure on copepods, hyperiids, and euphausiids was distributed relatively evenly because predatory plankton and nekton have different diets.

Isotope studies of aquatic organisms showed a comparatively uniform horizontal distribution of $\delta^{15}\text{N}$ values in dominant epipelagic zooplankton species across the Okhotsk Sea (Fig. 4), permitting construction of trophodynamic models for the entire sea.

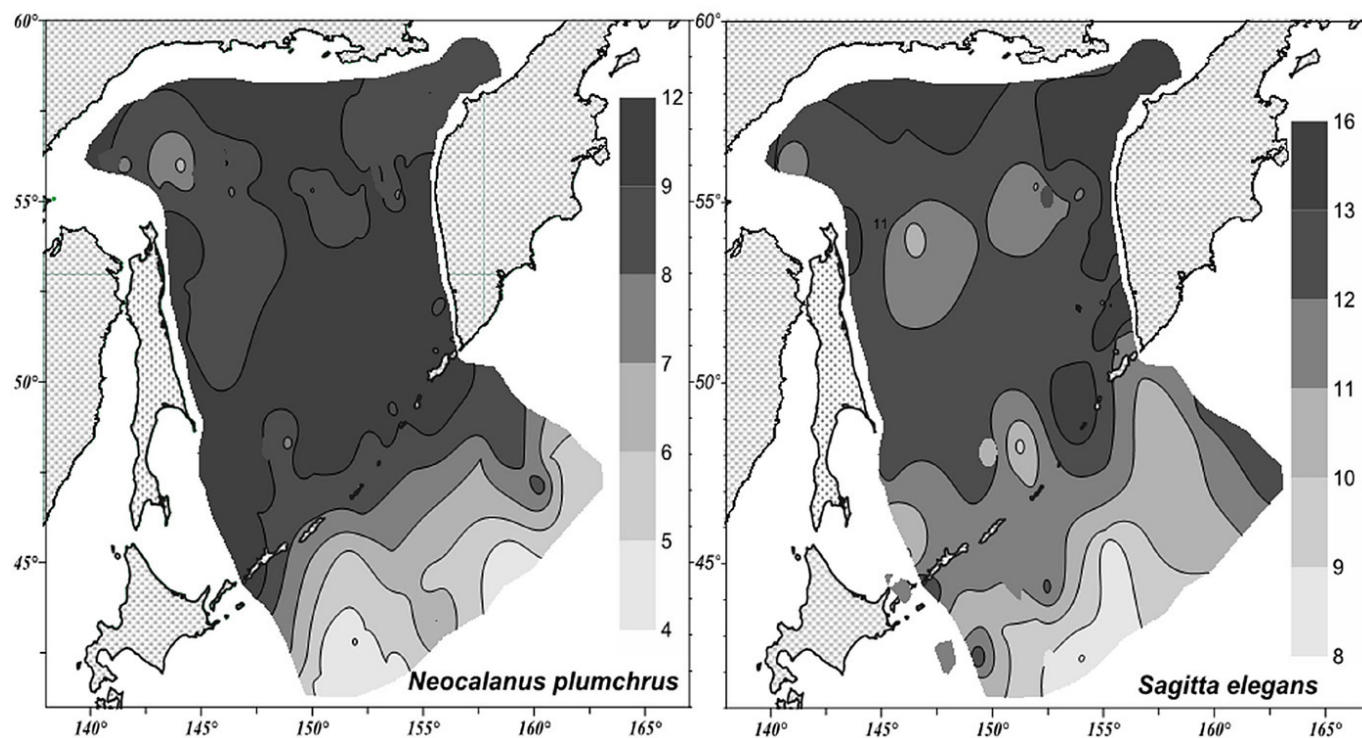


Figure 4. Horizontal distribution of $\delta^{15}\text{N}$ values in abundant non-predatory (*N. plumchrus*) and predatory (*S. elegans*) zooplankton in the Okhotsk Sea and adjacent Pacific waters.

English figure key: Left panel: *Neocalanus plumchrus*. Right panel: *Sagitta elegans*. Shading and adjacent numerical scales represent $\delta^{15}\text{N}$; latitude and longitude axes are retained unchanged.

In general, the major plankton groups show a regular increase in $\delta^{15}\text{N}$ values in all seasons, from non-predatory copepods to predatory jellyfish and chaetognaths (Fig. 5). Seasonal changes in zooplankton nitrogen isotope composition do not substantially alter trophic status: non-predatory plankton (filter feeders) remains at trophic level I, and predatory plankton at trophic level III. Mean $\delta^{15}\text{N}$ values in the Okhotsk Sea are $9.0 \pm 0.8\text{‰}$ for non-predatory zooplankton and $12.8 \pm 0.9\text{‰}$ for predatory zooplankton; mean $\delta^{13}\text{C}$ values are -20.9 ± 1.7 and $-19.8 \pm 1.8\text{‰}$, respectively [Gorbatenko, 2018].

Translation note: Trophic level I for filter-feeding plankton is retained as printed here; elsewhere the source assigns these organisms to level II.

Original page 151 (source PDF page 9)

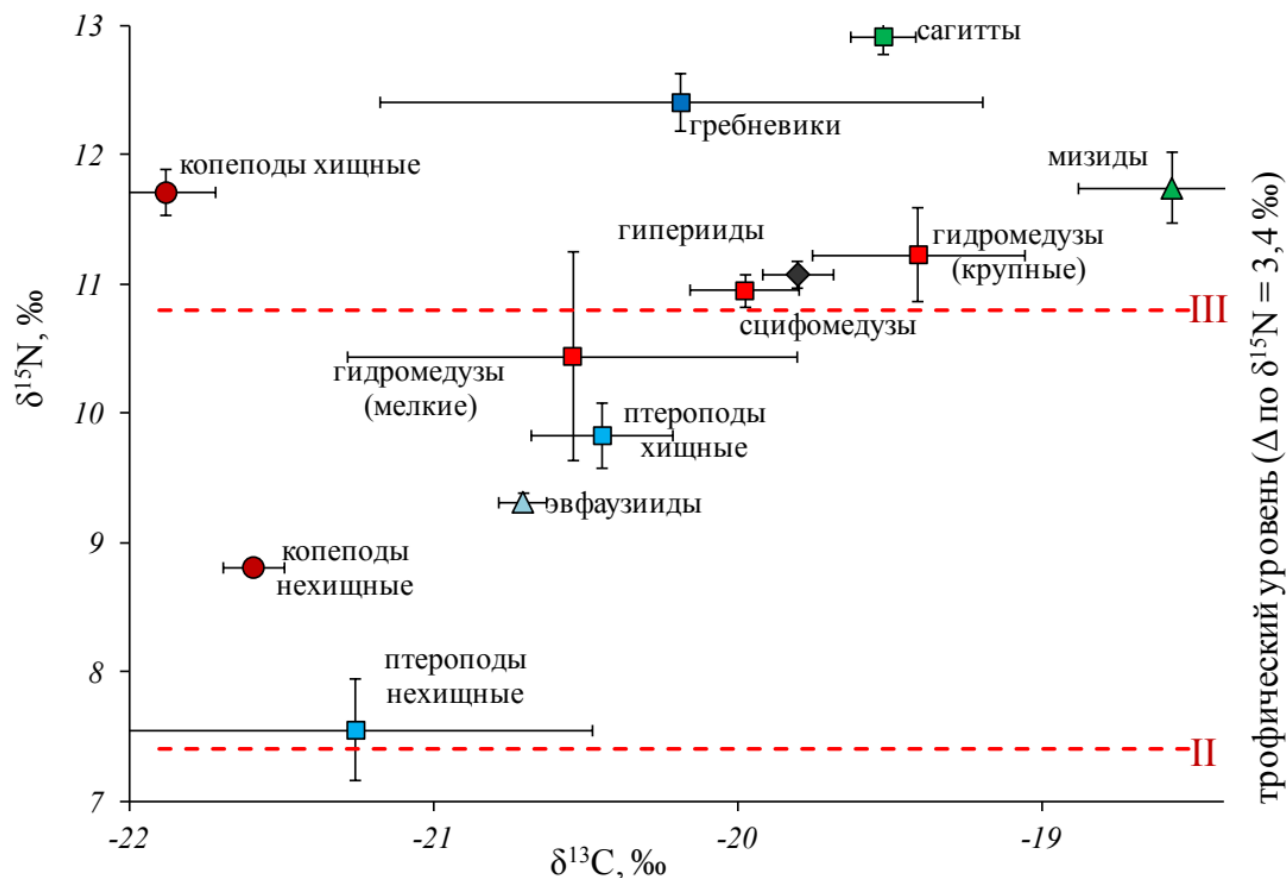


Figure 5. Relationships between stable carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopes in the major zooplankton groups of the Okhotsk Sea.

English figure key: Axes: horizontal $\delta^{13}\text{C}$ (‰), vertical $\delta^{15}\text{N}$ (‰). Right axis: trophic level (Δ in $\delta^{15}\text{N} = 3.4\text{‰}$); dashed lines II and III. English label key, by increasing vertical position: non-predatory pteropods; non-predatory copepods; euphausiids; predatory pteropods; small hydromedusae; scyphomedusae; hyperiids; large hydromedusae; predatory copepods; mysids; ctenophores; sagittas. Error bars and plotted positions remain as published.

Seasonal changes in isotope composition are also practically absent in the principal nekton species of the Okhotsk Sea, except for migrants from the Pacific Ocean [Gorbatenko et al., in press]. Annual mean isotope values can therefore be used in constructing trophodynamic models (Fig. 6).

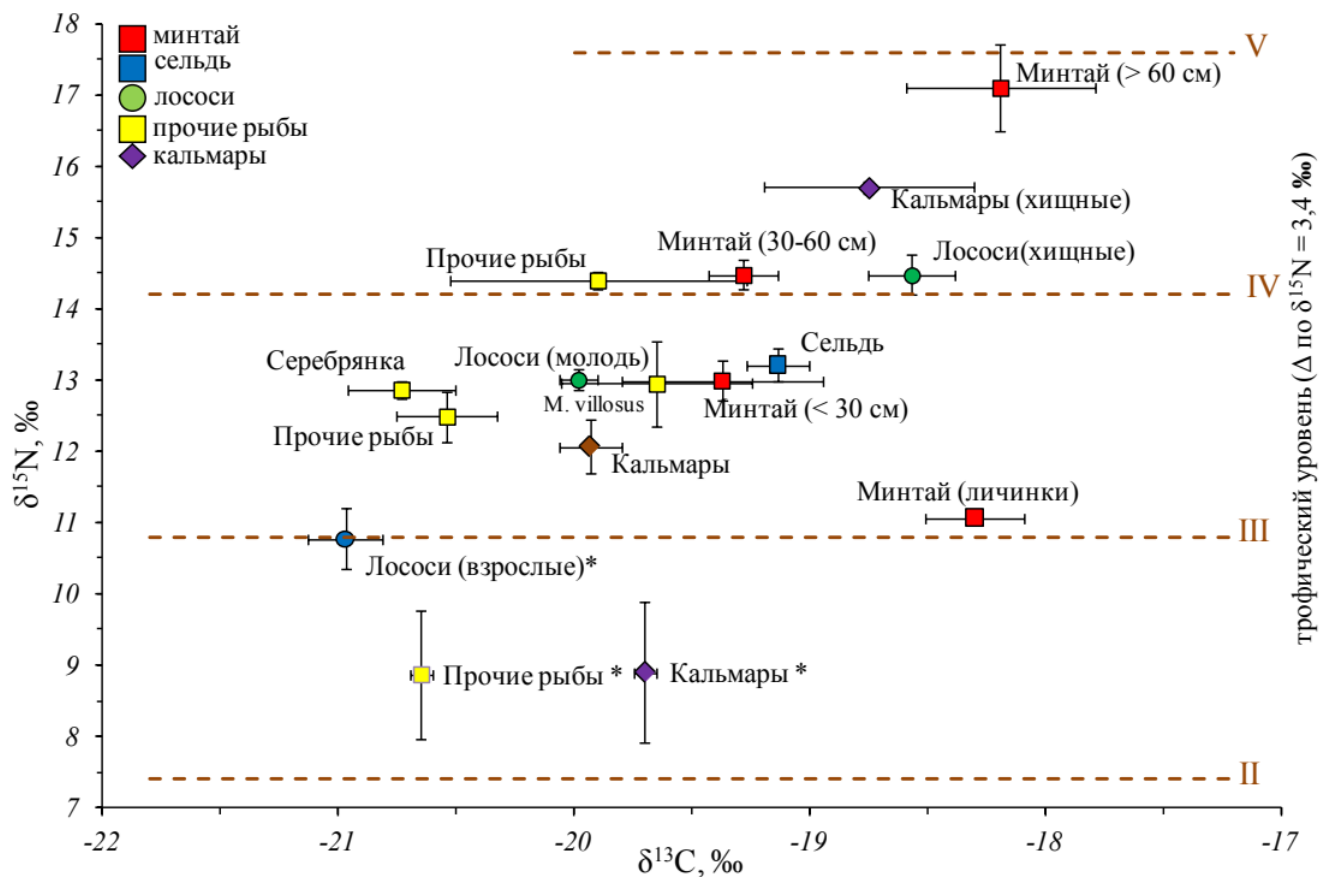


Figure 6. Relationships between stable carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopes in epipelagic nekton of the Okhotsk Sea, without separation by season. Asterisk: migrants from the Pacific Ocean.

English figure key: Axes: horizontal $\delta^{13}\text{C}$ (‰), vertical $\delta^{15}\text{N}$ (‰). Right axis: trophic level (Δ in $\delta^{15}\text{N} = 3.4\text{‰}$), II-V. Legend: pollock (red square); herring (blue square); salmon (green circle); other fish (yellow square); squid (purple diamond). Labels: pollock larvae; pollock <30 cm; pollock 30-60 cm; pollock >60 cm; herring; juvenile salmon; adult salmon*; predatory salmon; capelin (M. villosus); deep-sea smelt; other fish (several groups, including other fish*); squid; predatory squid; squid*. Asterisk denotes Pacific migrants.

Original page 152 (source PDF page 10)

Mean $\delta^{13}\text{C}$ values in the investigated fish species range from -20.2‰ in deep-sea smelt to -17.9‰ in very large pollock, over 60 cm, which leads a near-bottom existence. Nekton $\delta^{15}\text{N}$ values, excluding Pacific migrants, range from 11.0‰ in pollock larvae to 16.9‰ in pollock over 60 cm and reflect predation by individual species. Nitrogen isotope composition varies not only among species but also during ontogeny. Pollock trophic level changes quite substantially with growth (Fig. 6): $\delta^{15}\text{N}$ data show a clear relationship between changes in the trophic status of differently sized individuals and the proportions of plankton, benthos, and nekton in their diets [Gorbatenko and Levitskaya, 2016]. A similar lifetime shift to higher trophic levels is characteristic of many aquatic organisms, though the strength of this tendency varies. Unlike pollock, for example, the diet of Okhotsk Sea herring is based on two plankton groups, euphausiids and copepods, belonging to trophic level II and, to a lesser extent, III [Gorbatenko and Melnikov, 2016]. Because the qualitative composition of herring prey scarcely changes with age, carbon and nitrogen isotope composition remains constant [Gorbatenko et al., 2013].

Elemental composition studies indicate that organic carbon content in the wet weight of crustaceans (copepods, euphausiids, mysids, and hyperiids) is twice that of Oikopleura and sagittas, and more than an order of magnitude higher than in jellyfish (Fig. 7). A single biomass-to-carbon conversion factor should therefore not be used for all zooplankton in production calculations, as in some studies [Dvoretzky and Dvoretzky, 2010; and others], because zooplankton carbon content depends directly on the dominance of a particular group or its constituent species.

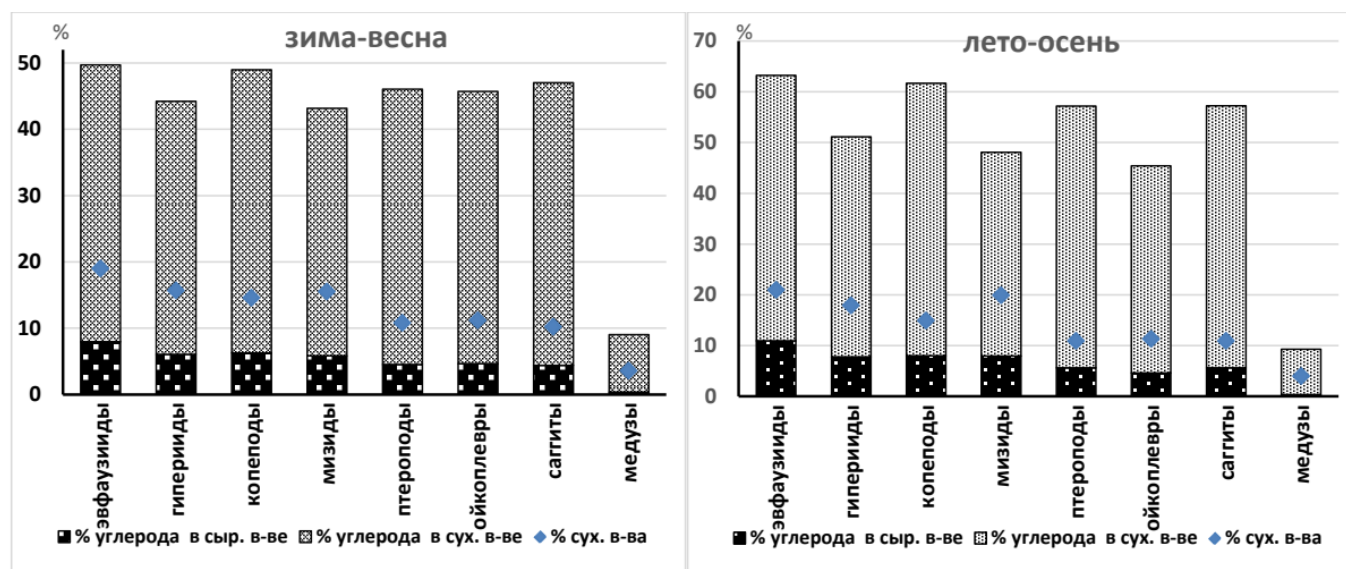


Figure 7. Organic carbon content of Okhotsk Sea zooplankton.

English figure key: Panels: winter-spring (left); summer-autumn (right). Vertical axes: %. Groups, left to right in each panel: euphausiids; hyperiids; copepods; mysids; pteropods; Oikopleura; sagittas; jellyfish. Legend: solid dark bars, carbon as % of wet matter; patterned bars, carbon as % of dry matter; blue diamonds, dry matter as % of wet matter.

Nekton organic carbon content varies from 43.1 to 61.1% of dry matter and from 7.2 to 18.4% of wet matter (Fig. 8).

Functioning of pelagic communities in 2000-2014

To determine the production potential of the principal elements of the Okhotsk Sea ecosystem, as in V. Ya. Berger's [2007] study of the White Sea ecosystem, individual constituents of production processes were analyzed and an attempt was made to relate them both to one another and to matter and energy consumption at different levels of food chains.

The information accumulated in the 2000s on the biomass and production of groups and components at the main trophic levels in the pelagic zone was used to calculate mean values of the principal production parameters of the pelagic community (Table 3). Energy indices are usually used to measure the intensity...

Original page 153 (source PDF page 11)

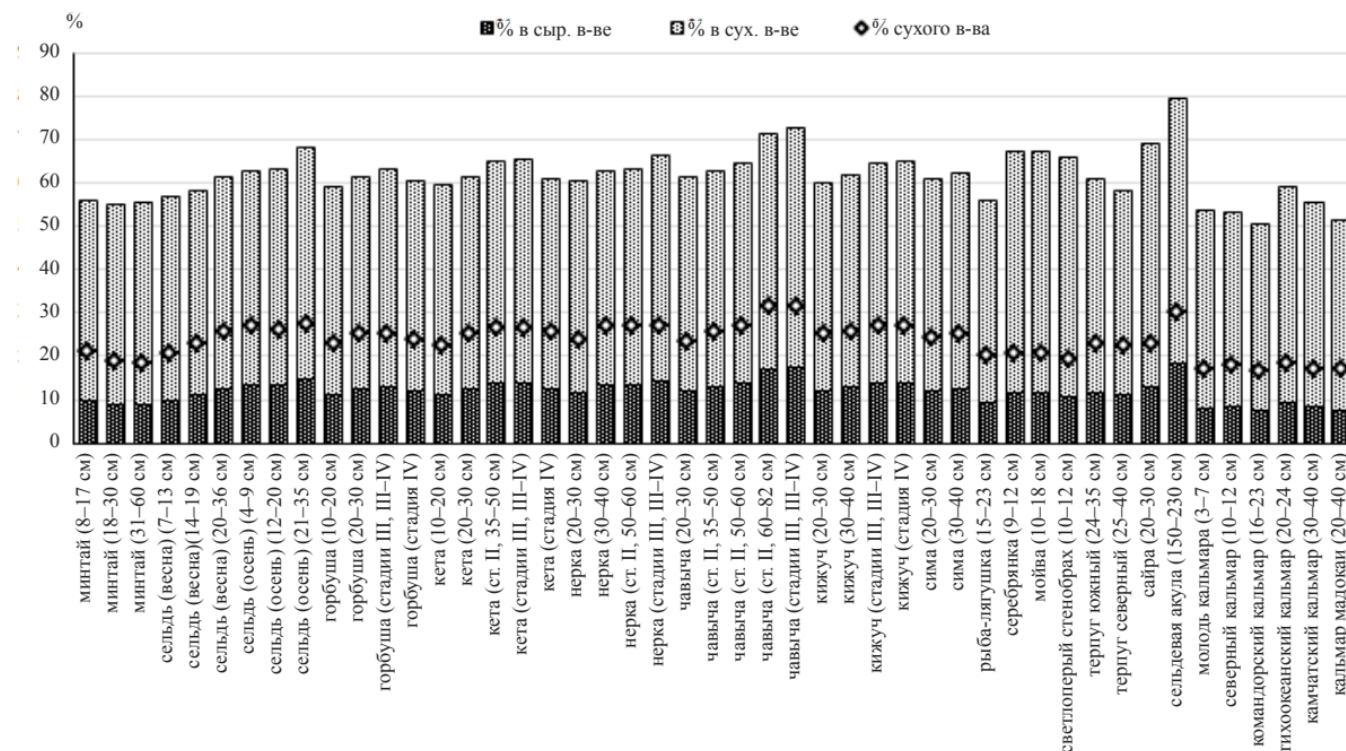


Figure 8. Organic carbon content of the dominant pelagic fish and squid species of the Okhotsk Sea.

English figure key: Vertical axis: %. Legend: dark bars, carbon as % of wet matter; patterned bars, carbon as % of dry matter; black diamonds, dry matter %. The ordered English category key below follows the original from left to right; III, III-IV, IV, and II are the stages printed in the original labels.

Figure 8: ordered English category key

No.	English category	No.	English category	No.	English category
1	Pollock (8-17 cm)	17	Chum salmon (stages III, III-IV)	33	Masu salmon (30-40 cm)
2	Pollock (18-30 cm)	18	Chum salmon (stage IV)	34	Frogfish (15-23 cm)
3	Pollock (31-60 cm)	19	Sockeye salmon (20-30 cm)	35	Deep-sea smelt (9-12 cm)
4	Herring, spring (7-13 cm)	20	Sockeye salmon (30-40 cm)	36	Capelin (10-18 cm)
5	Herring, spring (14-19 cm)	21	Sockeye salmon (stage II, 50-60 cm)	37	Light-finned Stenobranchius (10-12 cm)
6	Herring, spring (20-36 cm)	22	Sockeye salmon (stages III, III-IV)	38	Southern greenling (24-35 cm)
7	Herring, autumn (4-9 cm)	23	Chinook salmon (20-30 cm)	39	Northern greenling (25-40 cm)
8	Herring, autumn (12-20 cm)	24	Chinook salmon (stage II, 35-50 cm)	40	Pacific saury (20-30 cm)
9	Herring, autumn (21-35 cm)	25	Chinook salmon (stage II, 50-60 cm)	41	Salmon shark (150-230 cm)
10	Pink salmon (10-20 cm)	26	Chinook salmon (stage II, 60-82 cm)	42	Juvenile squid (3-7 cm)
11	Pink salmon (20-30 cm)	27	Chinook salmon (stages III, III-IV)	43	Boreal squid (10-12 cm)
12	Pink salmon (stages III, III-IV)	28	Coho salmon (20-30 cm)	44	Commander squid (16-23 cm)
13	Pink salmon (stage IV)	29	Coho salmon (30-40 cm)	45	Pacific squid (20-24 cm)
14	Chum salmon (10-20 cm)	30	Coho salmon (stages III, III-IV)	46	Kamchatka squid (30-40 cm)
15	Chum salmon (20-30 cm)	31	Coho salmon (stage IV)	47	Madokai squid (20-40 cm)
16	Chum salmon (stage II, 35-50 cm)	32	Masu salmon (20-30 cm)		

...of community functioning. In this case carbon equivalents were used (see Figs. 7 and 8). As noted above, phytoplankton production in the Okhotsk Sea, trophic level I, was estimated at 694.8 million tC/year, and microheterotroph production at 136.2 million tC/year. Their sum is 831.0 million tC/year of "primary food," or 151.8×10^9 million tonnes wet weight (Table 3).

Translation note: The wet-weight expression " 151.8×10^9 million tonnes" is preserved from the Russian source. Table 3 gives 15176.0 million tonnes for this total; the translation does not reconcile the difference.

Calculations based on Okhotsk Sea zooplankton biomass, production, and organic carbon content showed that filter-feeding copepod production at trophic level II is 117.2 million tC/year, representing 66% in carbon equivalents (Table 3). Our estimates [Gorbatenko, 2018] put long-term mean total annual prey consumption by copepods at 2945 million tonnes. Phytoplankton predominates in their annual diet, 76.8% by biomass; microheterotrophs account for 21.5%. Euphausiid production is estimated at 56.1 million tC/year, or 31.6% of trophic level II production in carbon equivalents (Table 3). Annual euphausiid food consumption is 1445.6 million tonnes; phytoplankton accounts for 78.9% of the diet and microheterotrophs for 19.4% of total annual food consumption [Gorbatenko, 2016b]. Together, the annual rations of the primary consumers, copepods and euphausiids, require 28.9% of total "primary food" production.

Energy transfer at trophic level III proceeds through predatory plankton (see Fig. 5) and nekton (Fig. 6). The dominant predatory plankton groups contribute most substantially, accounting for 96.5% of trophic level III in carbon equivalents (Table 3). Sagitta production is 12.78 million tC/year, or 69.8% of trophic level III production in carbon equivalents. Hyperiids and jellyfish account for 25.9 and 0.8%, respectively. K. M. Gorbatenko [2016a] showed that long-term mean total annual prey consumption by sagittas reaches 320 million tonnes. Their annual diet is dominated by copepods, 281.4 million tonnes, or 87.9% by biomass of total consumption, corresponding to 64.7% of gross copepod stock. Euphausiids account for only 2.2%, and sagittas for 2.0%. Standard consumption calculations showed that hyperiids consume 152.5 million tonnes of food annually. Copepods dominate the hyperiid diet, with annual consumption of 56.7 million tonnes, 13.0% of gross Okhotsk Sea copepod stock. In total, predatory plankton consumes 16.2%...

Original page 154 (source PDF page 12)

Table 3. Principal production parameters of the pelagic community of the Okhotsk Sea in 2000-2014.

B wet: biomass, million tonnes wet matter. Wet/C: wet-matter/carbon coefficient. B C*: biomass, million tC/year as printed. P/B C: P/B ratio in carbon. P wet: production, million tonnes wet matter. P C: production, million tC/year. P/area: production, gC/m² as printed. Share: component importance at its trophic level in carbon, %. TL: trophic level.

TL	Ecosystem component	B wet	Wet/C	B C*	P/B C	P wet	P C	P/area	Share %
I	Phytoplankton	63.2	20.0	3.2	220.0	13896.0	694.8	450.0	83.6
I	Microheterotrophs	64.0	9.4	6.8	20.0	1280.0	136.2	88.2	16.4
I	Total	127.2	12.8	10.0	83.4	15176.0	831.0	538.2	100
II	Euphausiids	58.7	10.6	5.6	10.10	592.8	56.1	36.3	31.6
II	Copepods	108.3	14.0	7.7	15.2	1645.7	117.2	75.9	66.0
II	Other	5.1	13.3	0.3	10.8	55.9	4.1	2.7	2.4
II	Total	172.1	12.6	13.6	13.0	2294.4	177.4	114.9	100
III	Hyperiid	5.90	14.50	0.41	11.70	69.03	4.752	3.10	25.90
III	Sagittas	42.70	20.0	2.13	6.0	256.20	12.780	8.30	69.80
III	Jellyfish	4.10	285.20	0.01	10.0	41.0	0.144	0.10	0.80
III	Herring	2.50	7.20	0.35	0.34	0.86	0.119	0.10	0.70
III	Deep-sea smelt	1.80	8.70	0.21	0.29	0.52	0.060	0.04	0.30
III	Capelin	0.80	8.80	0.09	0.35	0.26	0.030	0.02	0.20
III	Salmon	0.80	8.20	0.09	1.14	0.86	0.105	0.10	0.60
III	Squid	0.50	11.90	0.04	2.50	1.25	0.105	0.10	0.60
III	Pollock (<30 cm)	0.80	10.20	0.10	2.50	2.0	0.197	0.10	1.10
III	Other*	0.11	10.02	0.01	0.40	0.14	0.010	0.01	0.04
III	Total	60.01	17.55	3.44	5.36	372.12	18.302	11.97	100
IV	Pollock 30-60 cm	8.520	11.11	0.767	0.70	5.960	0.537	0.348	82.2
IV	Squid	0.500	11.94	0.042	2.50	1.250	0.105	0.068	16.0
IV	Predatory salmon	0.020	8.20	0.002	0.84	0.017	0.002	0.001	0.3
IV	Baleen whales	0.721	5.43	0.133	0.01	0.007	0.001	0.001	0.2
IV	Other**	0.150	9.09	0.017	0.50	0.075	0.008	0.005	1.3
IV	Total	9.911	10.32	0.961	0.68	7.309	0.653	0.423	100
V	Pollock >60 cm	0.500	11.11	0.045	0.20	0.100	0.009	0.006	61.0
V	Predatory mammals***	0.244	5.43	0.045	0.03	0.007	0.001	0.001	9.1
V	Other****	0.120	5.43	0.022	0.20	0.024	0.005	0.003	29.9
V	Total	0.864	7.72	0.112	0.13	0.131	0.015	0.010	100

* Other small fish. ** Predatory fish and seabirds. *** Toothed whales and pinnipeds. **** Sharks, daggertooth, lancetfish, and other predatory fish, including halibuts with pelagic feeding.

Translation note: the source labels carbon biomass with /year. This header and all printed values, including discrepancies with prose and Figure 9, have been retained.

...of zooplankton production, with the principal pressure on copepods, 19.7% (Table 3). The contribution of functional nekton groups is approximately an order of magnitude smaller: they account for only 3.4% of the total organic matter concentrated at trophic level III (Table 3).

Within nekton, juvenile pollock, herring, salmon, squid, and deep-sea smelt contribute most substantially to energy transformation. The remaining nekton, including young-of-the-year pollock, account for less than 0.1% and are assigned to "other."

Original page 155 (source PDF page 13)

After predatory zooplankton consumption is subtracted, the actual zooplankton production available to fish is 163.5 million tC/year, or 2230 million tonnes wet weight/year.

Herring

Herring predominated among nekton species at trophic level III; its average Okhotsk Sea stock in 2000-2014 was about 2.5 million tonnes [Gorbatenko and Melnikov, 2016]. V. I. Radchenko [1994] gives a mean herring P/B ratio of 0.34. Herring production is estimated at 0.86 million tonnes wet weight/year. Using data on nekton organic carbon, herring production is 0.119 million tC/year, or 0.7% of trophic level III in carbon equivalents (Table 3). Our studies showed that long-term mean total annual prey consumption by herring was 36.9 million tonnes [Gorbatenko and Melnikov, 2016]. Zooplankton predominated in the annual diet: 35.9 million tonnes, or 97.3% by biomass (4.2% of gross zooplankton stock).

Salmon

Pacific salmon inhabit the Okhotsk Sea for only part of the year. Long-term mean adult salmon biomass in summer 2000-2014 was 483.2 thousand tonnes; autumn biomass of juvenile and adult salmon was determined as 272.4 thousand tonnes (unpublished data of A. N. Starovoitov). The most abundant salmon species in the Far Eastern seas is pink salmon *Oncorhynchus gorbuscha*, which has the highest production potential among salmon. According to V. I. Radchenko [1994], the mean annual P/B ratio of pink salmon is 7.70, whereas that of chum salmon of different ages is 0.84. The average P/B ratio for differently sized pink, chum (*O. keta*), and sockeye (*O. nerka*) salmon in the Okhotsk Sea was 1.34; these species account for 98.6% of all salmon in the sea. Calculations showed that their total production in the Okhotsk Sea was 0.105 million tC/year, or 0.6% of total trophic level III production in carbon equivalents (Table 3).

Squid

Macroplankton is particularly important in the diet of small and medium-sized squid, while larger adults mainly eat nekton. Our data put total squid prey consumption at 32.4 million tonnes/year, with plankton averaging 64% of the diet and nekton 36%. Isotope studies place *Onychoteuthis borealijaponica*, *Berrytheuthis magister*, and *Gonatopsis borealis* at trophic level III; *Gonatus madokai*, *Gonatus onyx*, and others are at IV. Epipelagic squid grow rapidly, with annual P/B ratios estimated at 2.5-3.7, averaging 3.1 [Zuev et al., 1985; Nesis, 1985]. In 1979-2009, long-term mean nekton biomass in the Okhotsk Sea was estimated at 17.254 million tonnes, of which only about 2.0 million tonnes (1.2%) was squid [Shuntov and Temnykh, 2011]. Overall, cephalopod biomass in the Okhotsk Sea reaches 3 million tonnes and is concentrated principally in the bathypelagic and mesopelagic zones [Lapko, 1996]. As active migrants, squid ascend into the epipelagic layer to feed, and abundance there is particularly high at night. For calculating squid production in the epipelagic layer, we adopted a biomass of 1 million tonnes because these nekton organisms form concentrations in the epipelagic layer only at night. The mean P/B ratio for all squid is 2.5. Taking into account their organic carbon content, the lowest value among the nekton groups studied, their production is 0.105 million tC/year and their share of total trophic level III production is 0.6% in carbon equivalents (Table 3).

Deep-sea smelt

Eight mesopelagic fish species have been recorded in the Okhotsk Sea epipelagic layer, with deep-sea smelt predominating. Despite its high gross biomass, averaging 1.8 million tonnes annually, deep-sea smelt production at trophic level III does not exceed 0.06 million tC/year, or 0.3% in carbon equivalents (Table 3). Our data put its long-term mean total prey consumption at 10.29 million tonnes/year (see Fig. 3), or 5.3% of total nekton prey consumption. Euphausiids, 46.1%, and copepods, 40.7%, predominate in its annual diet.

Capelin

Capelin has the lowest production indices at trophic level III, with production of 0.03 million tC/year and a share of 0.2% in carbon...

Original page 156 (source PDF page 14)

...equivalents (Table 3). Despite its low production indices, capelin is exceeded only by pollock, herring, and squid in the volume of prey consumed in the Okhotsk Sea, accounting for 7.5% of total nekton food consumption (Fig. 3). Long-term mean total prey consumption by capelin is estimated at 14.74 million tonnes/year. Zooplankton predominates in its annual diet: 14.3 million tonnes, or 97.9% by weight.

Juvenile pollock (<30 cm)

Pollock up to 30 cm long are typical planktivores [Gorbatenko, 1997], and their Okhotsk Sea biomass in 2000-2014 was comparable with salmon and capelin, averaging 0.8 million tonnes [Ovsyannikov et al., 2013]. V. I. Radchenko [2015] gives a mean juvenile pollock P/B ratio of 2.5, close to that of squid. Juvenile pollock production is estimated at 2 million tonnes wet weight/year. Taking organic carbon content into account, production is 0.197 million tC/year, or 1.1% of trophic level III in carbon equivalents (Table 3). Thus, because of their high production, juvenile pollock make the largest contribution to total nekton production at trophic level III. Our data put their long-term mean total annual prey consumption at 16.5 million tonnes. Zooplankton predominates in the annual diet, 98.9% by biomass.

At trophic level IV in the Okhotsk Sea are differently sized pollock, 30-60 cm long, predatory salmon, squid, baleen whales, and seabirds. Among all nekton species, pollock has the leading role in the pelagic community; its biomass in the 2000s averaged 8.52 million tonnes wet weight [Ovsyannikov et al., 2013]. The total impact of 30-60 cm pollock on the biota, expressed as prey consumption, is 80.8 million tonnes/year, more than 50% of total nekton prey consumption [Gorbatenko and Levitskaya, 2016]. The principal consumption pressure falls on euphausiids, about 47 million tonnes/year, more than 20% of their gross stock. The mean annual pollock P/B ratio depends on the proportion of young individuals and varies from 0.6 to 0.8 [Dulepova and Sokolovsky, 1988; Shuntov et al., 1993; Dulepova, 2013]. Long-term mean production of this species in the 2000s is estimated at 0.54 million tC/year, equal to 82.2% of total aquatic organism production at trophic level IV (Table 3).

Pelagic squid feeding principally on nekton account for 16.9% of the energy concentrated at trophic level IV (Table 3). Predatory salmon account for about 2.4% of total Okhotsk Sea salmon biomass [Pelagic Macrofauna..., 2012]. Based on organic carbon content, predatory salmon production is 0.002 million tC/year and their share at trophic level IV is 0.3% (Table 3).

Baleen whale biomass in the Okhotsk Sea is estimated at 0.72 million tonnes wet weight (http://www.pices.int/publications/presentations/PICES_14/S3/Miyashita.pdf). The mean baleen whale P/B ratio is taken as 0.01, the same as for whales in Pacific waters around the Kuril Islands [Zavolokin, 2014]; production is 0.001 million tC/year (Table 3). The principal "other" organisms at trophic level IV are seabirds, whose Okhotsk Sea abundance is estimated at 15 million individuals [Shuntov, 2016].

The smallest energy quantity passes through trophic level V in the Okhotsk Sea: 0.015 million tC/year (Table 3). This level is based on fourth-order consumers at the apex of the food pyramid: very large pollock, sharks, and marine mammals (toothed whales and pinnipeds). Long-term mean very large pollock biomass in TINRO-Center surveys during the 2000s was 0.5 million tonnes; shark biomass was 0.1 million tonnes. Toothed whale biomass is estimated at 0.135 million tonnes (http://www.pices.int/publications/presentations/PICES_14/S3/Miyashita.pdf), and pinniped biomass at 0.109 million tonnes (unpublished data of A. E. Kuzin). This group has no rapidly growing forms, and its P/B ratio is only 0.11 (Table 3). For Okhotsk Sea mammals, as for...

Original page 157 (source PDF page 15)

...Pacific waters around the Kuril Islands, the P/B ratio is taken as 0.03 [Zavolokin, 2014], and their share at trophic level V is 9.1%. Organic carbon content in marine mammal wet weight, as in sharks, is 18.4%.

Based on the materials presented above, a generalized diagram of energy transformation in the Okhotsk Sea pelagic community was constructed (Fig. 9). It emphasizes the established understanding that the principal transfer of matter and energy in ecosystems occurs at low and intermediate trophic levels. The strongest energy flow is from phytoplankton to the most abundant group, copepods. Energy consumption by sagittas is also substantial at trophic level III, and by pollock at levels IV and V.

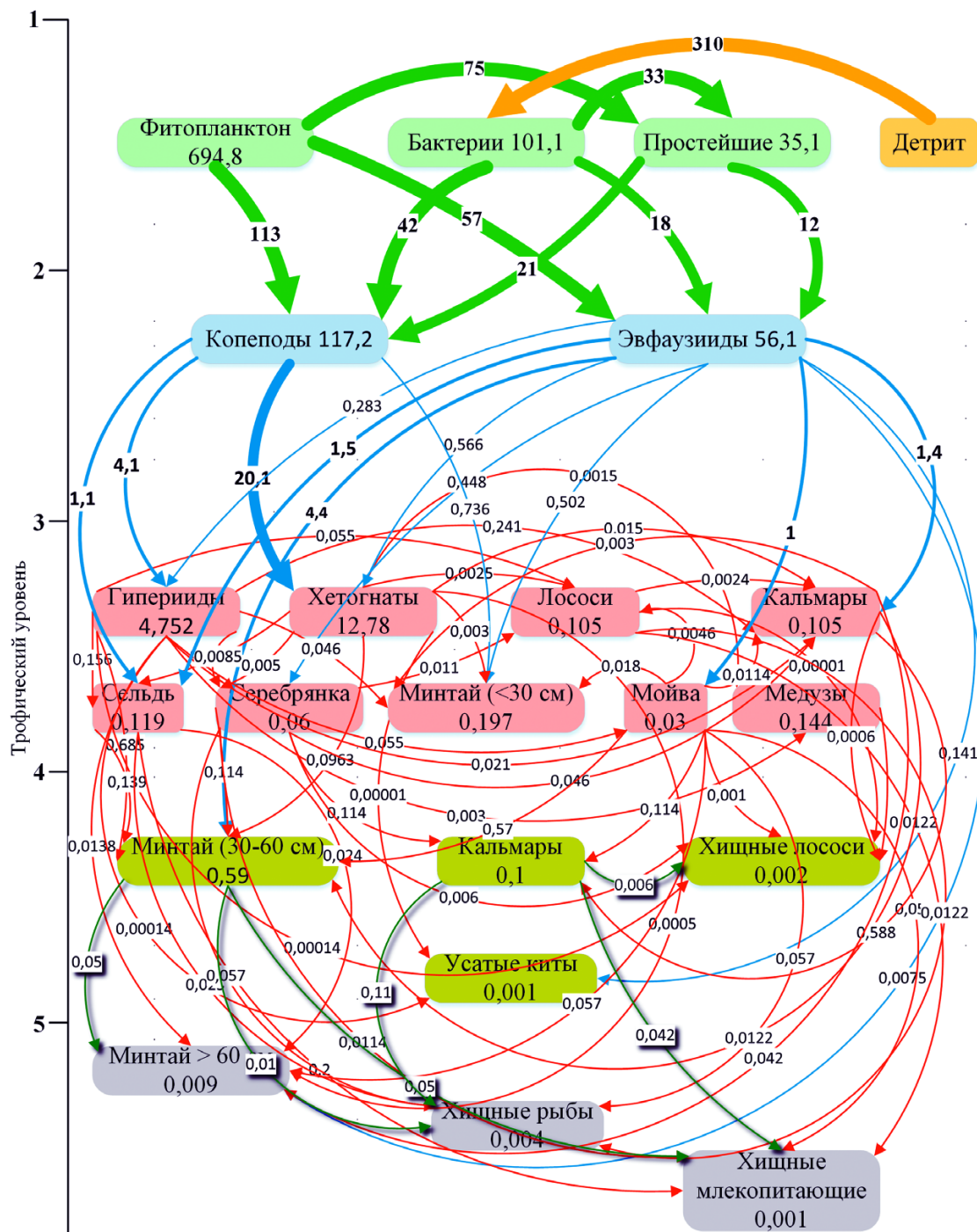


Figure 9. Energy flows in the pelagic community of the Okhotsk Sea in the 2000s. Numbers in boxes: production, million tC/year. Numbers on lines: the quantity of energy consumed by the next trophic link, million tC/year. Line-width classes, from thinnest to thickest: less than 1; 1-5; 5-10; 10-30; more than 30 million tC/year [Gorbatenko, 2018; with additions].

Figure 9: complete English node key

Production is in million tC/year, exactly as shown in the boxes. Detritus has no printed production value. Vertical axis: trophic levels I-V. Colours identify the original flow categories; arrows and numerical labels have not been altered or reinterpreted.

English node	Production	English node	Production
Phytoplankton	694.8	Deep-sea smelt	0.06
Bacteria	101.1	Pollock <30 cm	0.197
Protozoa	35.1	Capelin	0.03
Detritus	-	Jellyfish	0.144
Copepods	117.2	Pollock 30-60 cm	0.59
Euphausiids	56.1	Squid, level IV	0.1
Hyperiid	4.752	Predatory salmon	0.002
Chaetognaths	12.78	Baleen whales	0.001
Salmon	0.105	Pollock >60 cm	0.009
Squid, level III	0.105	Predatory fish	0.004
Herring	0.119	Predatory mammals	0.001

The published English caption describes the line numbers as "underlined numbers" and the next link as the "next higher trophic level." The Russian caption instead says numbers on the line and the subsequent trophic link; its wording is used in the translated caption above.

Total annual production of aquatic organisms in the Okhotsk Sea epipelagic layer is estimated at more than 1 billion tC (1027.4 million tC), or 17.85 billion tonnes wet weight (Fig. 10).

Original page 158 (source PDF page 16)

Excluding primary production, which accounts for 67.6% of gross production in carbon equivalents, the largest contributions to organic matter production are made by microheterotrophs (13.25%) and the dominant zooplankton groups: copepods (11.40%), euphausiids (5.46%), sagittas (1.24%), and hyperiids (0.46%). Contributions of various nekton groups, pollock, squid, salmon, and deep-sea smelt, are approximately one to two orders of magnitude smaller. The heterotrophic portion of the epipelagic community accounts for 32.4% of total production (Fig. 10).

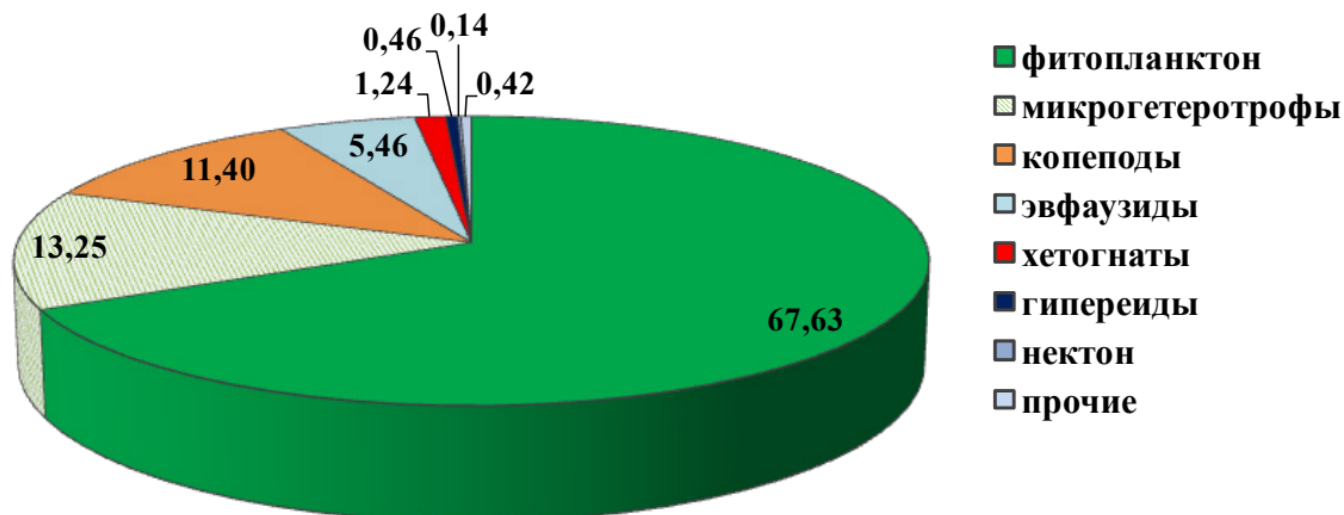


Figure 10. Shares of the principal pelagic community components in total annual organic matter production in the epipelagic layer of the Okhotsk Sea, 2000-2014, %.

English figure key: Legend and printed percentages: phytoplankton 67.63%; microheterotrophs 13.25%; copepods 11.40%; euphausiids 5.46%; chaetognaths 1.24%; hyperiids 0.46%; nekton 0.14%; other 0.42%. Caption heading in the chart: Share (%) of total annual production (1027.4 million tC/year) in the epipelagic layer of the Okhotsk Sea.

Conclusion

The Okhotsk Sea is characterized by abundant zooplankton development. According to the dominance of animal or plant food in their diets, zooplankton species form a system consisting of two functional elements: "non-predatory" zooplankton, first-order consumers, and "predatory" zooplankton, second- to third-order consumers. The identified trophic groupings are formed principally by these taxa: non-predatory zooplankton, copepods and euphausiids; predatory zooplankton, principally chaetognaths, hyperiids, and jellyfish.

Total zooplankton production for the entire sea in the 2000s, including non-predatory plankton production of 2275 million tonnes and predatory plankton production of 341 million tonnes, is estimated at 2616 million tonnes wet weight.

Predatory zooplankton consumed 16.2% of total annual zooplankton production, with the principal pressure on copepod populations, 19.7% of total annual copepod production. Nekton consumed 6.2% of annual zooplankton production, with maximum pressure on amphipods and euphausiids, whose consumption was one to two orders of magnitude greater than that by predatory zooplankton. Throughout the year, predatory zooplankton and nekton together consumed 22.4% of total annual zooplankton production.

Information accumulated in the 2000s on biomass, production, isotope composition, and organic carbon quantity in groups and components at the major pelagic trophic levels was used to calculate mean values of the principal epipelagic community production parameters. Annual organic matter production was 831.0 million tC at trophic level I, 177.400 at II, 18.100 at III, 0.740 at IV, and 0.016 million tC/year at V. Along the grazing food chain, organic matter is transferred from trophic level I through two principal taxonomic groups of first-order consumers, copepods and euphausiids, at trophic level II. The stronger energy flow, 66%, passes through copepods. At trophic level III the principal energy flow passes through predatory plankton, averaging 96.6%. Among nekton at trophic level III...

Original page 159 (source PDF page 17)

...pollock up to 30 cm predominate, accounting for 1.1% of the total energy passing through that level. At trophic levels IV and V, the principal energy flow passes through differently sized pollock over 30 cm, 82.2 and 61.0%, respectively.

Annual aquatic organism production in the Okhotsk Sea pelagic zone exceeds 1 billion tC (1027.2 million tC), or 17.8 billion tonnes wet weight. Excluding primary production, which accounts for 67.6% of gross production in carbon equivalents, the largest contributions are made by microheterotrophs, 13.3%, and the dominant zooplankton groups: copepods, 11.4%; euphausiids, 5.5%; sagittas, 1.2%; hyperiids, 0.5%. Contributions of functional nekton groups, pollock, herring, squid, salmon, and deep-sea smelt, are approximately one to two orders of magnitude smaller. The heterotrophic portion of Okhotsk Sea plankton communities accounts for about 32% of total production in those communities.

The strong development of representatives of trophic level II, zooplankton, principally copepods and euphausiids, suggests that pressure from planktonic and nektonic predators could be higher. In other words, the Okhotsk Sea food supply could potentially support a larger quantity of nekton than the present level, indicating opportunities to expand pasture aquaculture of valuable pelagic fish, including salmon.

The results contribute to the development of ideas about the patterns of functioning and dynamics of Okhotsk Sea pelagic communities.

Funding

The work received no special sponsorship funding.

Compliance with ethical standards

The authors declare no conflict of interest.

Information on author contributions

K. M. Gorbatenko: data collection; analysis and interpretation of results; preparation of the draft article. I. V. Melnikov: analysis and interpretation of results; final editing of the article.

Bibliography translated from the Russian list

Berger V. Ya. Production Potential of the White Sea. Monograph. St. Petersburg: Zoological Institute, Russian Academy of Sciences, 2007. 292 pp. (Investigations of the Fauna of the Seas, vol. 60(68)).

Vinberg G. G. General Foundations for the Study of Aquatic Ecosystems. Monograph. Leningrad: Nauka, 1979. 273 pp.

Vinogradov M. E., Shushkina E. A. Functioning of Plankton Communities in the Ocean Epipelagic Zone. Monograph. Moscow: Nauka, 1987. 240 pp.

Volkov A. F. Methods for collecting and processing plankton and nekton feeding samples (step-by-step instructions). Izvestiya TINRO, 2008, vol. 154, pp. 405-416.

Gorbatenko K. M. Distribution, biomass, and interannual dynamics of sagittas in the Okhotsk Sea. Izvestiya TINRO, 2016a, vol. 184, pp. 168-177.

Gorbatenko K. M. Quantitative distribution and feeding of euphausiids in the Okhotsk Sea. Izvestiya TINRO, 2016b, vol. 185, pp. 204-214.

Gorbatenko K. M. Composition, Structure, and Dynamics of Okhotsk Sea Plankton. Extended abstract of Candidate of Biological Sciences dissertation. Vladivostok: Institute of Marine Biology, Far Eastern Branch, Russian Academy of Sciences, 1997. 24 pp.

Gorbatenko K. M. Trophodynamics of Aquatic Organisms in the Okhotsk Sea. Extended abstract of Doctor of Biological Sciences dissertation. Vladivostok: Institute of Marine Biology, Far Eastern Branch, Russian Academy of Sciences, 2018. 48 pp.

Gorbatenko K. M., Kiyashko S. I., Lazhentsev A. E., et al. Investigation of the trophic structure of the Okhotsk Sea pelagic ecosystem from the nitrogen and carbon isotope composition of organic matter in aquatic organisms. Abstracts of the 17th Symposium on Stable Isotopes. Moscow: GEOKHI, 2004a, pp. 68-69.

Gorbatenko K. M., Merzlyakov A. Yu., Shershenkov S. Yu. Feeding characteristics of differently sized pollock larvae, *Theragra chalcogramma* (Pallas, 1814), on the western Kamchatka shelf. *Biologiya Morya*, 2004b, vol. 30, no. 2, pp. 131-137.

Original page 160 (source PDF page 18)

Bibliography translated from the Russian list (continued)

- Gorbatenko K. M., Lazhentsev A. E., Kiyashko S. I. Seasonal dynamics of the trophic status of Okhotsk Sea zooplankton (from stable C and N isotope analysis). *Izvestiya TINRO*, 2014, vol. 177, pp. 25-39.
- Gorbatenko K. M., Levitskaya A. V. Food composition, daily rations, and prey consumption volumes of pollock in the Okhotsk Sea in the 2000s. *Izvestiya TINRO*, 2016, vol. 185, pp. 194-203.
- Gorbatenko K. M., Melnikov I. V. Trophic studies of Okhotsk Sea herring in the 2000s (food composition, daily rations, and prey consumption assessment over the annual cycle). *Izvestiya TINRO*, 2016, vol. 185, pp. 185-193.
- Gorbatenko K. M., Melnikov I. V., Kiyashko S. I. Seasonal dynamics of fish and squid trophic status in the Okhotsk Sea pelagic zone (from stable isotope analysis of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$). *Voprosy Ikhtiologii*, in press.
- Gorbatenko K. M., Melnikov I. V., Kiyashko S. I., et al. Stock specificity and trophic status of herring in the pelagic zone of the northern Okhotsk Sea. *Izvestiya TINRO*, 2013, vol. 172, pp. 237-252.
- Dvoretzky V. G., Dvoretzky A. G. Distribution of zooplankton biomass in the Barents Sea in 2007. *Izvestiya TINRO*, 2010, vol. 161, pp. 162-171.
- Dulepova E. P. Use of the food supply by nekton during periods of high abundance in the Okhotsk Sea. *Izvestiya TINRO*, 2013, vol. 173, pp. 146-163.
- Dulepova E. P. Comparative Bioproductivity of Macroecosystems of the Far Eastern Seas. Monograph. Vladivostok: TINRO-Center, 2002. 274 pp.
- Dulepova E. P., Sokolovsky A. S. Pollock production in the Okhotsk Sea. Abstracts of the 3rd All-Union Conference on Marine Biology. Kyiv: Institute of Biology of the Southern Seas, 1988, part 2, p. 12.
- Zavolokin A. V. Food Availability for Pacific Salmon During Marine and Oceanic Feeding. Extended abstract of Doctor of Biological Sciences dissertation. Vladivostok: TINRO-Center, 2014. 48 pp.
- Zaika V. E. Comparative Productivity of Aquatic Organisms. Monograph. Kyiv: Naukova Dumka, 1983. 206 pp.
- Zaika V. E. Specific Production of Aquatic Invertebrates. Monograph. Kyiv: Naukova Dumka, 1972. 147 pp.
- Zuev G. V., Nigmatullin Ch. M., Nikolsky V. N. Nektonic Oceanic Squid. Monograph. Moscow: Agropromizdat, 1985. 223 pp.
- Ivanova M. B. Production of Planktonic Crustaceans in Fresh Waters. Monograph. Leningrad: Zoological Institute, USSR Academy of Sciences, 1985. 222 pp.
- Lapko V. V. Composition, Structure, and Dynamics of Nekton in the Okhotsk Sea Epipelagic Layer. Extended abstract of Candidate of Biological Sciences dissertation. Vladivostok: TINRO-Center, 1996. 24 pp.
- Pelagic Macrofauna of the Okhotsk Sea: Tables of Occurrence, Abundance, and Biomass, 1984-2009. V. P. Shuntov, L. N. Bocharov, I. V. Volvenko, V. V. Kulik; edited by V. P. Shuntov and L. N. Bocharov. Vladivostok: TINRO-Center, 2012. 800 pp.
- Matveev V. I. Hydrochemical Conditions of Biological Productivity in the Okhotsk Sea. Extended abstract of Candidate of Geographical Sciences dissertation. Vladivostok: TINRO-Center, 2006. 24 pp.
- Matveev V. I., Zhigalov I. A. Assessment of primary production in dynamically active zones of the Okhotsk Sea. *Voprosy Promyslovoy Okeanologii*, 2008, issue 5, no. 2, pp. 208-215.
- Naletova I. A., Sapozhnikov V. V., Metreveli M. P. Features of primary production distribution in summer and assessment of total production in the Okhotsk Sea. *Comprehensive Studies of the Okhotsk Sea Ecosystem*. Moscow: VNIRO, 1997, pp. 98-103.
- Nesis K. N. Oceanic Cephalopods: Distribution, Life Forms, Evolution. Monograph. Moscow: Nauka, 1985. 286 pp.
- Ovsyannikov E. E., Ovsyannikova S. L., Sheibak A. Yu. Dynamics and structure of pollock stocks in the northern Okhotsk Sea in the 2000s. *Izvestiya TINRO*, 2013, vol. 172, pp. 133-149.
- Radchenko V. I. The Ecopath model: a promising tool for comparative study of ecosystem functioning in the Okhotsk and Bering Seas. *Advanced Methods for Assessing Changes in Geophysical Phenomena, Ecosystems, and Technological Processes in the Study and Development of Natural Resources of the Subarctic Okhotsk Sea Region: Proceedings of the First Open International Conference*. Yuzhno-Sakhalinsk, 2011, pp. 70-81.
- Radchenko V. I. Composition, Structure, and Dynamics of Nekton Communities in the Bering Sea Epipelagic Layer. Extended abstract of Candidate of Biological Sciences dissertation. Vladivostok, 1994. 24 pp.
- Radchenko V. I. Characteristics of the Okhotsk Sea ecosystem inferred from modeling results. *Trudy VNIRO*, 2015, vol. 155, pp. 79-111.
- Sorokin Yu. I. Production of microflora. *Biological Productivity of the Ocean*. Moscow: Nauka, 1977, vol. 2, pp. 209-233.

Original page 161 (source PDF page 19)

Bibliography translated from the Russian list (continued)

- Sorokin Yu. I. The Black Sea: Nature, Resources. Monograph. Moscow: Nauka, 1982. 217 pp.
- Sorokin Yu. I., Sorokin P. Yu., Mamaeva T. I., Sorokina O. V. Bacterioplankton and planktonic ciliates in the Okhotsk Sea. Comprehensive Studies of the Okhotsk Sea Ecosystem. Moscow: VNIRO, 1997, pp. 210-216.
- Sorokin Yu. I., Sorokin P. Yu., Sorokina O. V., Mamaeva T. I. Primary production and heterotrophic microplankton in the Okhotsk Sea. Zhurnal Obshchey Biologii, 1995, vol. 56, no. 5, pp. 603-628.
- Sushchenya L. M. Quantitative Patterns of Crustacean Feeding. Monograph. Minsk: Nauka i Tekhnika, 1975. 207 pp.
- Chuchukalo V. I., Shebanova M. A., Dulepova E. P., Gorbatenko K. M. Life cycles and somatic production of euphausiids in the Okhotsk Sea. Izvestiya TINRO, 2013, vol. 173, pp. 164-183.
- Shebanova M. A. Production of certain abundant copepod species in the Okhotsk Sea during the summer-autumn period. Izvestiya TINRO, 2007, vol. 148, pp. 221-237.
- Shebanova M. A., Dulepova E. P., Chuchukalo V. I. Some biological features and production characteristics of Eucalanus bungii in the Okhotsk and Bering Seas. Izvestiya TINRO, 2010, vol. 162, pp. 281-293.
- Shebanova M. A., Chuchukalo V. I., Gorbatenko K. M. Life cycles and somatic production of hyperiids in the Okhotsk and Bering Seas. Izvestiya TINRO, 2014, vol. 176, pp. 155-176.
- Shebanova M. A., Chuchukalo V. I., Dulepova E. P. Some biological features and production characteristics of Oithona similis (Copepoda) in the Okhotsk Sea and western Bering Sea. Izvestiya TINRO, 2011, vol. 166, pp. 231-243.
- Shebanova M. A., Chuchukalo V. I., Dulepova E. P. Some biological features and production characteristics of Pseudocalanus newmani (Copepoda) in the Okhotsk and Bering Seas. Izvestiya TINRO, 2012, vol. 170, pp. 172-183.
- Shuntov V. P. Biology of the Far Eastern Seas of Russia. Monograph. Vladivostok: TINRO-Center, 2016, vol. 2. 604 pp.
- Shuntov V. P., Volkov A. F., Temnykh O. S., Dulepova E. P. Pollock in Ecosystems of the Far Eastern Seas. Monograph. Vladivostok: TINRO, 1993. 426 pp.
- Shuntov V. P., Temnykh O. S. Pacific Salmon in Marine and Oceanic Ecosystems. Monograph. Vladivostok: TINRO-Center, 2011, vol. 2. 473 pp.
- Shushkina E. A., Vinogradov M. E. Quantitative characteristics of the Pacific Ocean pelagic population: plankton biomass and production-destruction processes. Okeanologiya, 1988, vol. 28, issue 6, pp. 992-1000.
- Boysen-Jensen P. Valuation of the Limfjord. I Studies on the Fish-Food in the Limfjorg 1909-1917. Reports of the Danish Biological Station, 1919, vol. 26, pp. 1-44.
- Minagawa M., Wada E. Stepwise enrichment of ^{15}N along food chains: Further evidence and the relation between $\delta^{15}\text{N}$ and animal age. Geochimica et Cosmochimica Acta, 1984, vol. 48, issue 5, pp. 1135-1140. DOI: 10.1016/0016-7037(84)90204-7.
- Post D. M. Using stable isotopes to estimate trophic position: models, methods, and assumptions. Ecology, 2002, vol. 83, no. 3, pp. 703-718.

References as published in English in the source

- Berger, V.Ja., Produktionnyi potentsial Belogo morya (Production Potential of the White Sea), St. Petersburg: Zool. Inst. Ross. Akad. Nauk., 2007.
- Vinberg, G.G., Obshchie osnovy izucheniya vodnykh ekosistem (General Bases for the Study of Aquatic Ecosystems), Leningrad: Nauka, 1979.
- Vinogradov, M.Ye. and Shushkina, E.A., Funktsionirovaniye planktonnykh soobshchestv epipelagiali okeana (Functioning of Plankton Communities in the Epipelagic Zone of Ocean), Moscow: Nauka, 1987.
- Volkov, A.F., Technique of collecting and processing the samples of plankton and the samples on nekton feeding (step-by-step instructions), Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2008, vol. 154, pp. 405-416.
- Gorbatenko, K.M., Distribution, biomass, and year-to-year dynamics of Sagitta in the Okhotsk Sea, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2016a, vol. 184, pp. 168-177.
- Gorbatenko, K.M., Quantitative distribution and feeding of euphausiids in the Okhotsk Sea, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2016b, vol. 185, pp. 204-214.
- Gorbatenko, K.M., Composition, structure, and dynamics of plankton in the Sea of Okhotsk, Extended Abstract of Cand. Sci. (Biol.) Dissertation, Vladivostok: Inst. Biol. Morya, Dal'nevost. Otd. Ross. Akad. Nauk, 1997.
- Gorbatenko, K.M., Trophodynamics of aquatic organisms in the Sea of Okhotsk, Extended Abstract of Doctoral (Biol.) Dissertation, Vladivostok: Inst. Biol. Morya, Dal'nevost. Otd. Ross. Akad. Nauk, 2018. - 48 pp.

Original page 162 (source PDF page 20)

References as published in English in the source (continued)

- Gorbatenko, K.M., Kiyashko, S.I., Lazhentsev, A.Ye., Ignat'ev, A.V., and Velivetskaya, T.A., The study of trophic structure of the pelagic zone ecosystem of the Sea of Okhotsk, based on N and C isotope composition of organic matter of aquatic organisms, in Tezisy dokl. 17-go simpoziuma po stabil'nym izotopam (Proc. 17th Symp. on Stable Isotopes), Moscow: Inst. Geokhim. Anal. Khim., 2004a, pp. 68-69.
- Gorbatenko, K.M., Merzlyakov, A.Yu., and Shershenkov, S.Yu., Feeding patterns in different size larvae of walleye pollack, *Theragra chalcogramma* (Pallas, 1814) on the shelf of western Kamchatka, Russ. J. Mar. Biol., 2004b, vol. 30, no. 2, p. 113-120.
- Gorbatenko, K.M., Lazhentsev, A.E., and Kiyashko, S.I., Seasonal dynamics of trophic status for zooplankton in the Okhotsk Sea (on the data of testing the stable isotopes of C and N), Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2014, vol. 177, pp. 25-39.
- Gorbatenko, K.M. and Levitskaya, A.V., Food composition, daily rations, and volumes of consumption for walleye pollock in the Okhotsk Sea in the 2000s, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2016, vol. 185, pp. 194-203.
- Gorbatenko, K.M. and Melnikov, I.V., Trophic studies on the Okhotsk Sea herring in the 2000s (food composition, daily rations, assessment of consumption in the annual cycle), Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2016, vol. 185, pp. 185-193.
- Gorbatenko, K.M., Melnikov, I.V., Kiyashko, S.I. Seasonal dynamics of trophic status of fish and squid in the pelagic zone of the Sea of Okhotsk (according to the analysis of stable isotopes $d^{13}C$ and $d^{15}N$), Vopr. Ikhtiol. (in press).
- Gorbatenko, K.M., Melnikov, I.V., Kiyashko, S.I., Lazhentsev, A.E., and Aseeva, N.L., Population specificity and trophic status of herring in the pelagic layer of the northern Okhotsk Sea, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2013, vol. 172, pp. 237-252.
- Dvoretsky, V.G. and Dvoretsky, A.G., Zooplankton biomass distribution in the Barents Sea in 2007, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2010, vol. 161, pp. 162-171.
- Dulepova, E.P., Utilization of forage resources by nekton in periods of its high abundance in the Okhotsk Sea, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2013, vol. 173, pp. 146-163.
- Dulepova, E.P., Sravnitel'naya bioproduktivnost' makroekosistem dal'nevostochnykh morei (Comparative Bioproduktivnost' of Macroecosystems in Far Eastern Seas), Vladivostok: TINRO-Tsentr, 2002.
- Dulepova, E.P. and Sokolovskii, A.S., Production of Alaska pollock in the Sea of Okhotsk, in Tezisy dokl. 3-ei vses. konf. po morskoi biologii (Proc. 3rd All-Sov. Conf. Mar. Biol.), Kiev: Inst. Biol. Yuzhn. Morey, 1988, part. 2, pp. 12.
- Zavolokin, A.V., Food availability for Pacific salmon during the period of feeding in sea and ocean, Extended Abstract of Doctoral (Biol.) Dissertation, Vladivostok: TINRO-Tsentr, 2014.
- Zaika, V.E., Sravnitel'naya produktivnost' gidrobiontov (Comparative Productivity of Aquatic Organisms), Kiev: Naukova Dumka, 1983.
- Zaika, V.E., Udel'naya produktsiya vodnykh bespozvonochnykh (Specific Production of Aquatic Invertebrates), Kiev: Naukova Dumka, 1972.
- Zuev, G.V., Nigmatullin, Ch.M., and Nikol'skii, V.N., Nektonnye okeanicheskiye kal'mary (Nekton Oceanic Squid), Moscow: Agropromizdat, 1985.
- Ivanova, M.B., Produktsiya planktonnykh rakoobraznykh v presnykh vodakh (Production of Plankton Crustaceans in Fresh Waters), Leningrad: Zool. Inst., Akad. Nauk. SSSR, 1985.
- Lapko, V.V., Composition, structure, and dynamics of the epipelagic nekton in the Sea of Okhotsk, Extended Abstract of Cand. Sci. (Biol.) Dissertation, Vladivostok: TINRO-Tsentr, 1996.
- Shuntov, V.P., Bocharov, L.N., Volvenko, I.V., and Kulik, V.V., Makrofauna pelagialy Okhotskogo morya: tablitsy vstrechaemosti, chislennosti i biomassy. 1984-2009 (Pelagic Macrofauna of the Okhotsk Sea: Occurrence, Abundance, and Biomass. 1984-2009), Shuntov, V.P. and Bocharov, L.N., Eds., Vladivostok: TINRO-Tsentr, 2012.
- Matveev, V.I., Hydrochemical conditions of biological productivity of the Sea of Okhotsk, Extended Abstract of Cand. Sci. (Geogr.) Dissertation, Vladivostok: TINRO-Tsentr, 2006.
- Matveev, V.I., and Zhigalov, I.A., Estimation of the size of primary production of dynamically active zones in the Sea of Okhotsk, Vopr. Promysl. Okeanol., 2008, vol. 5, no. 2, pp. 208-215.
- Naletova, I.A., Sapozhnikov, V.V., and Metreveli, M.P., The particularities of the distribution of primary production in summer and the estimation of the total production in the Sea of Okhotsk, in Kompleksnye issledovaniya ekosistemy Okhotskogo morya (Complex Studies of Ecosystem of the Sea of Okhotsk), Moscow: VNIRO, 1997, pp. 98-103.
- Nesis, K.N., Okeanicheskiye golovonogiye molluski: rasprostraneniye, zhiznennyye formy, evolutsiya (Oceanic Cephalopods: Distribution, Life Forms, and Evolution), Moscow: Nauka, 1985.
- Ovsyannikov, E.E., Ovsyannikova, S.L., and Sheybak, A.Yu., Dynamics and structure of walleye pollock stock in the northern Okhotsk Sea in the 2000s, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2013, vol. 172, pp. 133-149.

Original page 163 (source PDF page 21)

References as published in English in the source (continued)

- Radchenko, V.I., The Ecopas model, a promising tool for comparative studying the functioning of the ecosystems in the Sea of Okhotsk and the Bering Sea, in Mater. Pervoi otkrytoi mezhdunar. konf. "Perspektivnye metody otsenki izmenenii geofizicheskikh yavlenii, ekosistem i tekhnologicheskikh protsessov pri izuchenii i osvoenii prirodnikh resursov subarkticheskogo Okhotomor'ya (Proc. 1st Open Int. Conf. "Advanced Assessment methods of Changing in Geophysics, Ecosystems and Technological Processes in Study and Development of Natural Resources of the Subarctic Okhotomor'e") (Yuzhno-Sakhalinsk, Russia, April 05-06, 2011), Yuzhno-Sakhalinsk, 2011, pp. 70-81.
- Radchenko, V.I., Composition, structure, and dynamics of nekton communities in the epipelagic layer of the Bering Sea, Extended Abstract of Cand. (Biol.) Sci. Dissertation, Vladivostok: TINRO-Tsentr, 1994.
- Radchenko, V.I., Characteristics of the Sea of Okhotsk ecosystem inferred from results of modeling, Tr. Vseross. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2015, vol. 155, pp. 79-111.
- Sorokin, Yu.I., Production of microflora, in Biologicheskaya produktivnost' okeana (Biological Productivity of the Ocean), Moscow: Nauka, 1977, vol. 2, pp. 209-233.
- Sorokin, Yu.I., Chernoye more: priroda, resursy (The Black Sea: The Nature and the Resources), Moscow: Nauka, 1982.
- Sorokin, Yu.I., Sorokin, P.Yu., Mamaeva, T.I., and Sorokina, O.V., The Sea of Okhotsk bacterioplankton and planktonic ciliates, in Kompleksnye issledovaniya ekosistemy Okhotskogo morya (Complex Studies of Ecosystem of the Sea of Okhotsk), Moscow: VNIRO, 1997, pp. 210-216.
- Sorokin, Yu.I., Sorokin, P.Yu., Sorokina, O.V., and Mamaeva, T.I., The primary production and planktonic microheterotrophs in the Okhotsk Sea, Zh. Obshch. Biol., vol. 56, no. 5, pp. 603-628.
- Sushchenya, L.M., Kolichestvennye zakonomernosti pitaniya rakoobraznykh (Quantitative Patterns of Feeding in Crustaceans), Minsk: Nauka i Tekhnika, 1975.
- Chuchukalo, V.I., Shebanova, M.A., Dulepova, E.P., and Gorbatenko, K.M., Life cycles and somatic production of euphausiids in the Okhotsk Sea, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2013, vol. 173, pp. 164-183.
- Shebanova, M.A., Production of some mass species of Copepoda in the Okhotsk Sea in summer-autumn, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2007, vol. 148, pp. 221-237.
- Shebanova, M.A., Dulepova, E.P., and Chuchukalo, V.I., Some features of Eucalanus bungii biology and its production characteristics in the Okhotsk and Bering Seas, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2010, vol. 162, pp. 281-293.
- Shebanova, M.A., Chuchukalo, V.I., and Gorbatenko, K.M., Life cycles and somatic production of hyperiids in the Okhotsk and Bering Seas, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2014, vol. 176, pp. 155-176.
- Shebanova, M.A., Chuchukalo, V.I., and Dulepova, E.P., Some features of biology and production parameters of Oithona similis from the Okhotsk and western Bering Seas, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2011, vol. 166, pp. 231-243.
- Shebanova, M.A., Chuchukalo, V.I., and Dulepova, E.P., Some biological features and productive parameters of Pseudocalanus newmani (Copepoda) in the Okhotsk and Bering Seas, Izv. Tikhookean. Nauchno-Issled. Inst. Rybn. Khoz. Okeanogr., 2012, vol. 170, pp. 172-183.
- Shuntov, V.P., Biologiya dal'nevostochnykh morei Rossii (Biology of the Far Eastern Seas of Russia), Vladivostok: TINRO-Tsentr, 2016, vol. 2.
- Shuntov, V.P., Volkov, A.F., Temnykh, O.S., and Dulepova, E.P., Mintai v ekosistemakh dal'nevostochnykh morei (Walleye Pollock in Ecosystems of the Far Eastern Seas), Vladivostok: TINRO-Tsentr, 1993.
- Shuntov, V.P. and Temnykh, O.S., Tikhookeanskii losos v morskikh i okeanicheskikh ekosistemakh (Pacific Salmon in Sea and Ocean Ecosystems), Vladivostok: TINRO-Tsentr, 2011, vol. 2.
- Shushkina, E.A. and Vinogradov, M.E., Quantitative characteristics of the population in the pelagic zone of the Pacific Ocean: Plankton biomass and production-destruction processes, Okeanologiya, 1988, vol. 28, no. 6, pp. 992-1000.
- Boysen-Jensen, P., Valuation of the Limfjord. I Studies on the fish-food in the Limfjorg 1909-1917, Rep. Danish Biol. Stn., 1919, vol. 26, pp. 1-44.
- Minagawa, M. and Wada, E., Stepwise enrichment of ^{15}N along food chains: Further evidence and the relation between $\delta^{15}\text{N}$ and animal age, Geochim. Cosmochim. Acta, 1984, vol. 48, no. 5, pp. 1135-1140. doi 10.1016/0016-7037(84)90204-7
- Post, D.M., Using stable isotopes to estimate trophic position: models, methods, and assumptions, Ecology, 2002, vol. 83, no. 3, pp. 703-718.

Received by the editorial office: 13 August 2018.

After revision: 25 October 2018.

Accepted for publication: 15 January 2019.